

AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies

*Dissertation submitted in partial fulfillment of the requirements
for the award of Degree of*

BACHELOR OF COMPUTER APPLICATIONS

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DEPARTMENT OF COMPUTER APPLICATIONS

SCHOOL OF COMPUTING

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Certificate

This is to certify that the project report entitled **“AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies”** is the bonafide work carried out and submitted by

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DECLARATION

We, here by declare that, the project entitled “**AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies**” developed by us at **MOHAN BABU UNIVERSITY, Tirupati** during the Academic year 2025-2026 and submitted to the Department of Computer Applications, MOHAN BABU UNIVERSITY for partial fulfilment for the award of Bachelor of Computer Applications (BCA).

We also declare that, the project is resulted by our own effort and that it has not been copied from anyone and not been submitted by anybody in any of the University or Institution or Research Centre.

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ABSTRACT

Mango (*Mangifera indica* L.) is among the most valuable kinds of fruits species across the globe and very significant in nutritional and economical terms, particularly to the tropical and subtropical farmers. Nonetheless, the serious biotic and abiotic stresses are at the moment threatening the cultivation of mango; namely, the leaf diseases and nutrient deficiencies, which may lead to immense losses in the yield. The traditional methods of diagnosis are mostly subjective, time-consuming, and inaccessible to the small-scale farmers. This paper thus dwells on this crucial matter by presenting an automated and deep learning based diagnostic utility on mango leaf health examination. The suggested tool uses DenseNet121 convolutional neural network that uses transfer learning to effectively detect eight major diseases, including Bacterial Canker, powdery Mildew, Anthracnose, Gall Midge, Sooty Mould, Die Back, Cutting Weevil, and nutrient deficiencies. The resulting model is trained with a balancing dataset of 4,000 high-resolution leaf images and it has an optimal state-of-the-art validation of 98.75%. The optimal computational complexity to be used on mobile implementation is attained. This study, as such, will contribute greatly to precision agriculture by offering expedient, precise and informative diagnostic tests thereby bridging the gap between the advanced artificial intelligence and the actual agricultural operations.

Keywords: densenet-121, mango leaf disease detection, nutrient deficiency classification, deep learning, computer vision.

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Short Bio-data of the Student

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1. INTRODUCTION

1.1 Introduction

Mango (*Mangifera indica* L.), popularly known as the “King of Fruits,” is one of the most important tropical fruit crops cultivated worldwide. India is the largest producer and exporter of mangoes, contributing significantly to the agricultural economy, rural employment, and nutritional security. The fruit is valued for its rich taste, high vitamin content, and commercial importance both in domestic and international markets. However, mango cultivation faces serious challenges due to various leaf diseases and nutrient deficiencies. These issues adversely affect photosynthesis, reduce plant vigour, cause premature leaf drop, poor fruit set, and lower fruit quality, ultimately leading to substantial yield losses. Leaf diseases such as Anthracnose, Powdery Mildew, Bacterial Canker, Die-back, Sooty Mould, and Red Rust (algal leaf spot) are common in mango orchards. In addition, nutrient deficiencies, particularly of Nitrogen (N), Potassium (K), Zinc (Zn), Iron (Fe), Boron (B), and Magnesium (Mg), produce visible symptoms on leaves including chlorosis, scorching of margins, interveinal yellowing, reduced leaf size, and leathery texture. These conditions make the plant more susceptible to diseases and further degrade overall productivity. Traditional detection and diagnosis of these problems largely depend on manual visual inspection by farmers or agricultural experts. This conventional method is highly subjective, time-consuming, labour-intensive, and often inaccurate, especially in large orchards or remote areas where access to experts is limited. Symptoms frequently become noticeable only after considerable damage has already occurred, resulting in delayed intervention, excessive use of chemical pesticides and fertilizers, increased production costs, and potential environmental hazards. In recent years, advancements in Artificial Intelligence (AI), Machine Learning (ML), Image Processing, and Computer Vision have opened new possibilities for automated, early, and accurate detection of plant health issues through leaf image analysis. This project proposes an intelligent AI-based system for the detection and classification of common mango leaf diseases and nutrient deficiencies. The system employs image preprocessing techniques such as resizing, normalization, noise reduction, and colour space conversion to handle variations in lighting and field conditions. Leaf segmentation is performed to separate the leaf from the background, followed by extraction of relevant features including colour histograms, texture features (using GLCM and LBP), and shape descriptors. These features are then analysed using traditional machine learning algorithms such as Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Additionally, the project explores deep learning models, particularly

DenseNet-121 with transfer learning, for improved classification accuracy. By shifting from reactive manual diagnosis to a proactive, automated, and intelligent monitoring approach, this system aims to enable farmers to identify problems at an early stage, take timely corrective actions, minimize yield losses, optimize resource utilization, and promote sustainable mango cultivation practices.

1.2 Aim of the Project:

The primary aim of this project is to design, develop, and implement an automated intelligent system for the detection and classification of mango leaf diseases and nutrient deficiencies using image processing and machine learning techniques. Specifically, the project aims to:

- Develop a system that can automatically detect and classify major mango leaf diseases (such as Anthracnose, Powdery Mildew, Bacterial Canker, etc.) and nutrient deficiencies (Nitrogen, Potassium, Zinc, Iron, etc.) from leaf images captured under real field conditions.
- Apply appropriate image preprocessing steps including resizing, normalization, noise reduction, and colour space conversion to improve image quality and handle variations in lighting, background, and field environments.
- Perform leaf segmentation to isolate the mango leaf from complex backgrounds, thereby enhancing feature extraction accuracy.
- Extract significant hand-crafted features such as colour histograms, texture features using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP), and shape descriptors for traditional machine learning models.
- Implement and compare multiple machine learning classifiers including Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) for accurate classification.
- Explore deep learning approaches, particularly DenseNet-121 with transfer learning, to achieve higher classification performance.
- Design a user-friendly interface that allows farmers or users to upload leaf images and receive instant detection results along with recommended remedial measures.

1.3 Project Domain

This project belongs to the interdisciplinary domain of Precision Agriculture and Smart Farming, with a strong focus on the application of Artificial Intelligence and Computer Vision in plant pathology and nutrient management. It specifically falls under the following domains:

- **Artificial Intelligence and Machine Learning:** Application of traditional ML algorithms (Random Forest, SVM, KNN) and deep learning models (DenseNet-121 with transfer learning) for automated classification of plant health issues.
- **Computer Vision and Image Processing:** Techniques such as image preprocessing, segmentation, feature extraction (colour, texture, and shape), and pattern recognition from leaf images.
- **Agricultural Technology:** Development of automated systems for early detection of mango leaf diseases and nutrient deficiencies to support precision nutrient and disease management.
- **Web or Mobile Application Development (optional extension):** Design of a simple interface for image upload and result visualization.

The project integrates knowledge from Agriculture (mango pathology and nutrition), Computer Science (machine learning and computer vision), and Software Engineering (system implementation). It addresses real-world challenges faced by mango farmers by applying cutting-edge AI technologies, making it highly relevant to the emerging field of AI-enabled Smart Agriculture and Sustainable Crop Management in India.

1.4 Scope of the Project

The scope of this project encompasses the development of an AI-based system capable of detecting and classifying major mango leaf diseases and nutrient deficiencies from static leaf images captured using smartphones or digital cameras under field conditions. The system will focus on the following key aspects:

- Preprocessing of input leaf images through resizing, normalization, noise reduction, and suitable colour space conversion.
- Segmentation of mango leaves from complex backgrounds to improve analysis accuracy.
- Extraction of hand-crafted features (colour histograms, GLCM, LBP texture, and shape descriptors) and their use in traditional machine learning models (Random Forest, SVM, KNN).
- Exploration of a deep learning model (DenseNet-121 with transfer learning) for enhanced performance.
- Classification into common disease categories (Anthracnose, Powdery Mildew, Bacterial Canker, etc.) and nutrient deficiency categories (Nitrogen, Potassium, Zinc, Iron, etc.).
- Generation of basic recommendations for corrective actions based on the detected issue.

The models will be trained and evaluated using a custom or publicly available dataset of annotated mango leaf images. All backend processing, including model training and inference, will be carried out using Python. A simple web or desktop interface may be developed for user interaction and result visualization. The scope is limited to static leaf images and does not include real-time video processing, pest detection on fruits, or integration with IoT sensors at this stage. The system aims to serve as a practical decision-support tool for mango farmers, extension workers, and agricultural researchers.

2. LITERATURE REVIEW

The application of Artificial Intelligence, Machine Learning, and Computer Vision techniques for automated plant disease and nutrient deficiency detection has gained significant attention in recent years. These technologies offer non-destructive, rapid, and accurate alternatives to traditional manual inspection methods, enabling early intervention and supporting precision agriculture practices. A substantial body of research has focused on various crops, with several studies specifically addressing mango leaf diseases and nutrient deficiencies. Early approaches primarily relied on hand-crafted feature extraction combined with traditional machine learning classifiers. Researchers extracted colour, texture, and shape features from leaf images and applied algorithms such as Support Vector Machine (SVM), K-Nearest Neighbours (KNN), Random Forest, and Artificial Neural Networks (ANN) for classification. These methods demonstrated reasonable accuracy in controlled laboratory conditions but often suffered from limited robustness when applied to images captured under real-field variations in lighting, background clutter, and leaf orientation. In the context of mango crops, several studies have explored the detection of common leaf diseases. Works on Anthracnose and Powdery Mildew detection have utilized colour histogram analysis and texture descriptors such as Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP). Some researchers combined these features with SVM and achieved classification accuracies ranging from 80% to 90% on relatively small datasets. However, performance tended to degrade when dealing with multiple disease classes or images with complex backgrounds. Nutrient deficiency detection in mango leaves has also been investigated using image processing techniques. Studies focusing on Nitrogen, Potassium, Zinc, and Iron deficiencies have employed colour space transformations (RGB to HSV or Lab) and statistical texture analysis to identify symptoms such as chlorosis, leaf scorching, and interveinal yellowing. Machine learning models like Random Forest and Decision Trees were frequently used, showing promising results in distinguishing deficiency symptoms from healthy leaves and certain diseases. Nevertheless, overlapping visual symptoms between certain nutrient deficiencies and early-stage diseases remained a major challenge, often leading to misclassification. With the advent of deep learning, Convolutional Neural Networks (CNNs) have largely outperformed traditional feature-based approaches in plant disease identification. Transfer learning using pre-trained models such as VGG16, ResNet50, InceptionV3, and DenseNet has become increasingly popular due to their ability to learn hierarchical features directly from raw images, reducing the dependency on manual feature engineering. Several studies reported classification accuracies exceeding 95% on datasets containing multiple plant

species and disease classes when using these architectures. Specifically for mango leaf analysis, recent research has applied deep learning models including CNN variants and Vision Transformers for disease classification. Some works fine-tuned DenseNet and EfficientNet architectures on mango leaf datasets and achieved improved generalization across varying field conditions. The use of data augmentation techniques such as rotation, flipping, brightness adjustment, and contrast enhancement has further helped in addressing the problem of limited dataset size and improving model robustness. Hybrid approaches combining traditional machine learning with deep learning features have also been explored. In these systems, deep CNNs are used as feature extractors, while classifiers like SVM or Random Forest perform the final decision-making. Such hybrid models have shown balanced performance in terms of accuracy and computational efficiency, making them suitable for deployment on resource-constrained devices. Despite notable progress, several gaps remain in the existing literature. Most studies focus either on disease detection or nutrient deficiency detection separately, with limited work addressing both problems simultaneously in a single system. Many models are trained on images captured in controlled environments and exhibit reduced performance on real-field images with complex backgrounds, shadows, and overlapping leaves. Additionally, very few systems provide actionable recommendations or integrate a user-friendly interface suitable for farmers with minimal technical expertise. The majority of research also lacks comprehensive comparison between traditional machine learning and deep learning approaches on the same mango leaf dataset. The proposed system addresses these limitations by developing a complete pipeline that handles both mango leaf diseases and nutrient deficiencies. It combines image preprocessing and segmentation techniques with traditional machine learning algorithms (Random Forest, SVM, KNN) based on hand-crafted features, while also exploring the deep learning model DenseNet-121 with transfer learning for improved accuracy. The system aims to deliver reliable performance under real-field conditions and provide a practical tool for early detection and management in mango cultivation.

3. PROBLEM DEFINITION

3.1 Existing System

Traditional methods for detecting mango leaf diseases and nutrient deficiencies primarily rely on manual visual inspection by farmers or agricultural extension officers. In these systems, growers physically examine the leaves for visible symptoms such as spots, lesions, powdery growth, yellowing, scorching, or abnormal texture. Based on their experience, they identify the problem and decide on appropriate remedial measures like pesticide application or fertilizer dosage. While this approach has been practiced for decades, it suffers from several critical limitations. Manual inspection is highly time-consuming and labour-intensive, making it difficult to monitor large orchards frequently. The accuracy of diagnosis heavily depends on the knowledge and subjectivity of the observer, often leading to inconsistent results and misidentification, especially during early stages of infection or when multiple problems exhibit similar symptoms. Moreover, manual methods lack scalability and are significantly affected by environmental factors such as poor lighting, shadows, overlapping leaves, and variations in leaf orientation. In recent years, some semi-automated systems based on basic image processing techniques or earlier machine learning models have been introduced. However, most of these systems focus on detecting only a limited number of diseases or deficiencies, suffer from lower accuracy in real-field conditions, and do not integrate both disease and nutrient deficiency detection in a single framework. They also lack robust preprocessing for field variations, comprehensive feature analysis, and a user-friendly interface for practical use by farmers. Very few systems provide actionable recommendations or automated alerts for timely intervention. The existing systems, therefore, fall short in providing a fast, accurate, reliable, scalable, and end-to-end solution for early detection of mango leaf diseases and nutrient deficiencies. These drawbacks highlight the need for an advanced automated system that can overcome the inefficiencies of manual and semi-automated approaches through the integration of image processing and state-of-the-art machine learning techniques.

3.2 Hardware Requirements

The successful implementation and execution of the proposed mango leaf disease and nutrient deficiency detection system require the following hardware components:

- Processor — Intel Core i3 or Higher
- Hard Disk — 500 GB SSD
- Keyboard — Standard Windows Keyboard
- Mouse — Two or Three Button Mouse
- Monitor — SVGA / HD Monitor
- RAM — 8 GB (Minimum) / 16 GB (Recommended)

a. Processor: The Intel Core i3 or higher processor is suitable for running the AI-based mango leaf detection system. It efficiently handles image preprocessing, feature extraction, and inference using trained machine learning and deep learning models. While model training may be performed on Google Colab with GPU acceleration, a standard i3/i5 processor is sufficient for development, testing, and real-time inference on pre-trained models.

b. Hard Disk (500 GB SSD): A minimum of 500 GB SSD provides adequate storage space for the operating system, Python environment, required libraries, mango leaf dataset, trained model weights, and project files. SSD is recommended for faster data loading during image processing and inference.

c. Keyboard & Mouse: Standard input devices are required for writing and editing Python code, developing the user interface, and navigating through the development environment.

d. Monitor (SVGA / HD Monitor): A standard monitor is necessary to comfortably view high-resolution leaf images, feature visualizations, classification results, and the interactive dashboard.

e. RAM (8 GB Minimum / 16 GB Recommended): A minimum of 8 GB RAM is required for smooth operation during inference and running the application. 16 GB RAM is recommended for better performance when processing multiple images or training deep learning models.

3.3 Software Requirements

The proposed system has been developed using the following software tools and technologies:

Category	Tools / Technologies	Description
Operating System	Windows 10 / 11 (64-bit)	Provides a stable environment for development and deployment with full compatibility for all required tools and libraries.
Programming Language	Python 3.8 or Higher	Core language used for implementing image processing, machine learning, and deep learning components.
Machine Learning Libraries	Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib	Used for data handling, preprocessing, visualization, and implementing ML algorithms like Random Forest, SVM, and KNN.
Deep Learning Framework	TensorFlow / Keras (DenseNet-121)	Enables building and training deep learning models using transfer learning for accurate mango leaf disease detection.
Development Environment	Google Colab, VS Code, PyCharm	Colab is used for training with GPU support, while VS Code/PyCharm are used for coding, debugging, and system integration.
Image Processing Tools	OpenCV, Pillow	Used for image preprocessing, segmentation, and feature extraction.
Dataset Annotation Tools	Roboflow, LabelImg	Helps in labeling and managing datasets efficiently for model training.

Table 3.3:Software Requirements

4. PROPOSED SYSTEM

4.1 Objectives

The primary objective of this proposed system is to develop an automated intelligent system for the detection and classification of mango leaf diseases and nutrient deficiencies using image processing and machine learning techniques. The specific objectives are as follows:

- To design a system that can automatically detect and classify major mango leaf diseases (such as Anthracnose, Powdery Mildew, Bacterial Canker, Die-back, Sooty Mould, and Red Rust) and nutrient deficiencies (Nitrogen, Potassium, Zinc, Iron, Boron, and Magnesium) from leaf images captured under real field conditions.
- To apply suitable image preprocessing techniques including resizing, normalization, noise reduction, and colour space conversion to handle variations in lighting, background, and environmental conditions.
- To perform effective leaf segmentation to isolate the mango leaf from complex backgrounds, thereby improving the accuracy of subsequent feature extraction and classification.
- To extract relevant hand-crafted features such as colour histograms, texture features using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP), and shape descriptors for use with traditional machine learning models.
- To implement and compare multiple machine learning classifiers including Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) for accurate classification of diseases and deficiencies.
- To explore and evaluate a deep learning approach using DenseNet-121 with transfer learning to achieve higher classification performance and better generalization.
- To develop a user-friendly interface that allows users (farmers or agricultural officers) to upload mango leaf images and receive instant detection results along with basic remedial recommendations.

4.2 Problem Formulation

Mango cultivation plays a vital role in India's agricultural economy, yet it suffers heavy losses due to leaf diseases and nutrient deficiencies. These problems reduce photosynthesis, weaken the plant, lower fruit quality, and cause significant yield reduction if not identified and treated at an early stage. Traditional manual inspection is slow, subjective, and often inaccurate, while existing semi-automated systems lack robustness under real-field conditions and do not simultaneously address both diseases and nutrient deficiencies in one framework. To overcome these challenges, this research proposes an AI-based mango leaf disease and nutrient deficiency detection system. The proposed solution follows a structured workflow as shown in the figure below. The system begins with the collection of mango leaf images representing healthy leaves, various diseases, and nutrient deficiency symptoms. During preprocessing, images undergo resizing, normalization, noise reduction, and colour space conversion to manage field variations. Leaf segmentation is performed to isolate the region of interest from the background. Relevant features including colour histograms, GLCM texture, LBP texture, and shape descriptors are extracted. These features are fed into traditional machine learning classifiers (Random Forest, SVM, KNN) for classification. In parallel, a deep learning model based on DenseNet-121 with transfer learning is explored for improved performance. The system finally generates the predicted class (disease or deficiency type) along with basic remedial suggestions. This comprehensive pipeline aims to provide faster, more accurate, and actionable detection compared to conventional methods, enabling timely intervention by mango farmers.

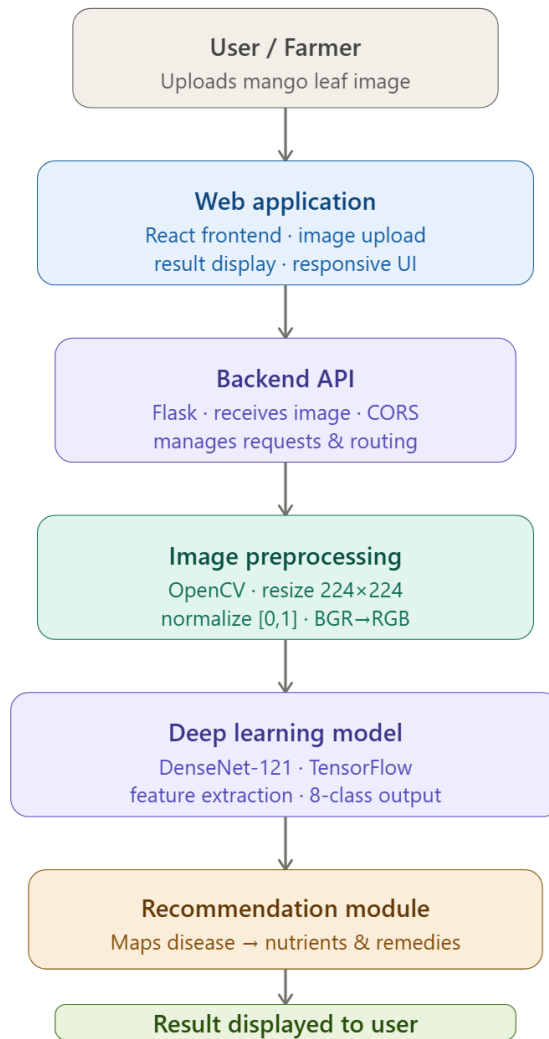


Fig 4.1: Workflow diagram

4.3 Methodology of Proposed System

The methodology of the proposed AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies follows a systematic end-to-end workflow, from image acquisition to final classification and recommendation. The detailed workflow of the proposed system is as follows:

1. **Image Acquisition:** High-resolution mango leaf images are collected under real field conditions using smartphones or digital cameras, covering healthy leaves, multiple disease types, and various nutrient deficiency symptoms.
2. **Image Preprocessing:** Input images are resized to a uniform dimension, normalized to bring pixel values to a standard range, subjected to noise reduction filters, and converted to appropriate colour spaces (such as HSV or Lab*) to minimize the effects of varying lighting and background conditions.
3. **Leaf Segmentation:** Segmentation techniques are applied to separate the mango leaf from the complex background, ensuring that only the relevant leaf area is used for further analysis and reducing interference from soil, other leaves, or non-leaf objects.
4. **Feature Extraction:** Multiple types of features are extracted from the segmented leaf images, including:
 - Colour histograms (to capture colour variations due to chlorosis or lesions),
 - Texture features using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP),
 - Shape descriptors (to identify changes in leaf structure or margin damage).
5. **Classification using Machine Learning:** The extracted features are fed into traditional machine learning algorithms — Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). These classifiers are trained and compared to determine the most suitable model for distinguishing between healthy leaves, different diseases, and nutrient deficiencies.
6. **Deep Learning Approach:** In addition to traditional models, a deep learning model

based on DenseNet-121 with transfer learning is implemented and evaluated. This model learns hierarchical features directly from the preprocessed images, potentially offering higher accuracy and better generalization.

- 7. Result Generation and Recommendation:** The system outputs the predicted class (specific disease or nutrient deficiency) along with a confidence score and provides basic remedial recommendations (e.g., suggested fungicide for Anthracnose or fertilizer application for Potassium deficiency).
- 8. User Interface:** A simple and intuitive interface (web or desktop-based) is developed to allow easy image upload and clear visualization of detection results.

This hybrid methodology ensures a balance between interpretability (through traditional ML) and high performance (through deep learning), making the system practical and effective for real-world mango orchard monitoring.

5. DATASET COLLECTION

5.1 Dataset Description

The performance of any machine learning or deep learning-based system for plant disease detection heavily depends on the quality, diversity, and size of the training dataset. In this research, a publicly available Mango Leaf Disease Dataset has been utilized for training, validation, and testing of the proposed models. This dataset consists of high-resolution mango leaf images captured under real-field conditions in mango-growing regions. It contains a total of approximately 4000 images (with around 1800 distinct leaves and the rest augmented through zooming and rotation). The images are labeled into eight classes, which include both common diseases and a healthy category:

- Anthracnose
- Bacterial Canker
- Cutting Weevil
- Die Back
- Gall Midge
- Healthy
- Powdery Mildew
- Sooty Mould

Each image is provided in JPG format with a resolution of 240×320 pixels. The dataset reflects realistic orchard conditions, including variations in lighting, background clutter (soil, other leaves, shadows), leaf orientation, and disease severity stages. This diversity helps the model learn robust features and generalize effectively to images captured by farmers using smartphones in real-world environments. Before feeding the images into the models, several preprocessing steps were applied to ensure consistency and improve training stability:

- Resizing of all images to 224×224 pixels (to match the input requirement of the DenseNet-121 architecture).
- Normalization of pixel intensities to the range $[0, 1]$.
- Colour space conversion from BGR to RGB.
- Data augmentation techniques such as rotation, flipping, zooming, and brightness/contrast adjustment (applied during training to artificially expand the dataset and reduce overfitting).

The dataset was split into training, validation, and testing sets (typically 70:15:15 ratio) to enable proper model evaluation. The well-balanced representation of the eight classes, including the “Healthy” category, allows the system to reliably distinguish between healthy leaves and various disease conditions, as well as identify overlapping symptoms that may relate to nutrient stress. This dataset serves as a strong benchmark for evaluating both traditional machine learning models (using hand-crafted features) and the deep learning model (DenseNet-121 with transfer learning). By utilizing this realistic and publicly accessible dataset, the proposed system achieves reliable detection and classification of mango leaf diseases, supporting early intervention by farmers.

6. SYSTEM DESIGN

6.1 High level Design Documentation

The high-level design of the proposed AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies follows a modular architecture that integrates image processing techniques, traditional machine learning models, and deep learning approaches. The system is designed to provide an end-to-end solution for mango leaf analysis, starting from image input to disease classification and generation of remedial recommendations. At the core of the system is a hybrid approach that combines hand-crafted feature extraction with both traditional machine learning classifiers and a deep learning model. The architecture consists of the following major modules:

1. **User Interface Layer:** A simple and intuitive web-based interface (developed using Flask) that allows users to upload mango leaf images and view detection results. It displays the predicted disease class, confidence score, and recommended remedies.
2. **Preprocessing Module:** This module handles image resizing to 224×224 pixels, colour space conversion (BGR to RGB), normalization of pixel values to the [0, 1] range, and basic noise reduction to improve image quality under varying field conditions.
3. **Leaf Segmentation Module (optional enhancement):** Techniques are applied to isolate the mango leaf from complex backgrounds, reducing interference from non-leaf elements such as soil or other vegetation.
4. **Feature Extraction Module:** For traditional machine learning models, relevant hand-crafted features are extracted, including colour histograms, texture features using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP), and shape descriptors.
5. **Classification Module:**
 - Traditional machine learning classifiers (Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)) are implemented using the extracted features.
 - A deep learning model based on DenseNet-121 with transfer learning is used for direct image classification, leveraging its dense connectivity to capture rich hierarchical features.
6. **Recommendation Module:** Based on the predicted disease class, the system provides nutrient-related suggestions and practical remedies (e.g., fungicide application, pruning, or fertilizer recommendations) to support timely farmer intervention.
7. **Backend Processing Layer:** All model training, inference, and image processing operations

are executed using Python scripts. The Flask framework serves as the backend server for handling image uploads and returning JSON responses with predictions and recommendations.

The system follows a layered architecture where the Flask web framework acts as the presentation and API layer, while the machine learning and deep learning models form the intelligent core. This design ensures high modularity, scalability, maintainability, and ease of deployment. The hybrid methodology balances interpretability (through traditional ML) with high accuracy (through DenseNet-121), making the system practical for real-world use by mango farmers.

6.1.1. Use-case diagram

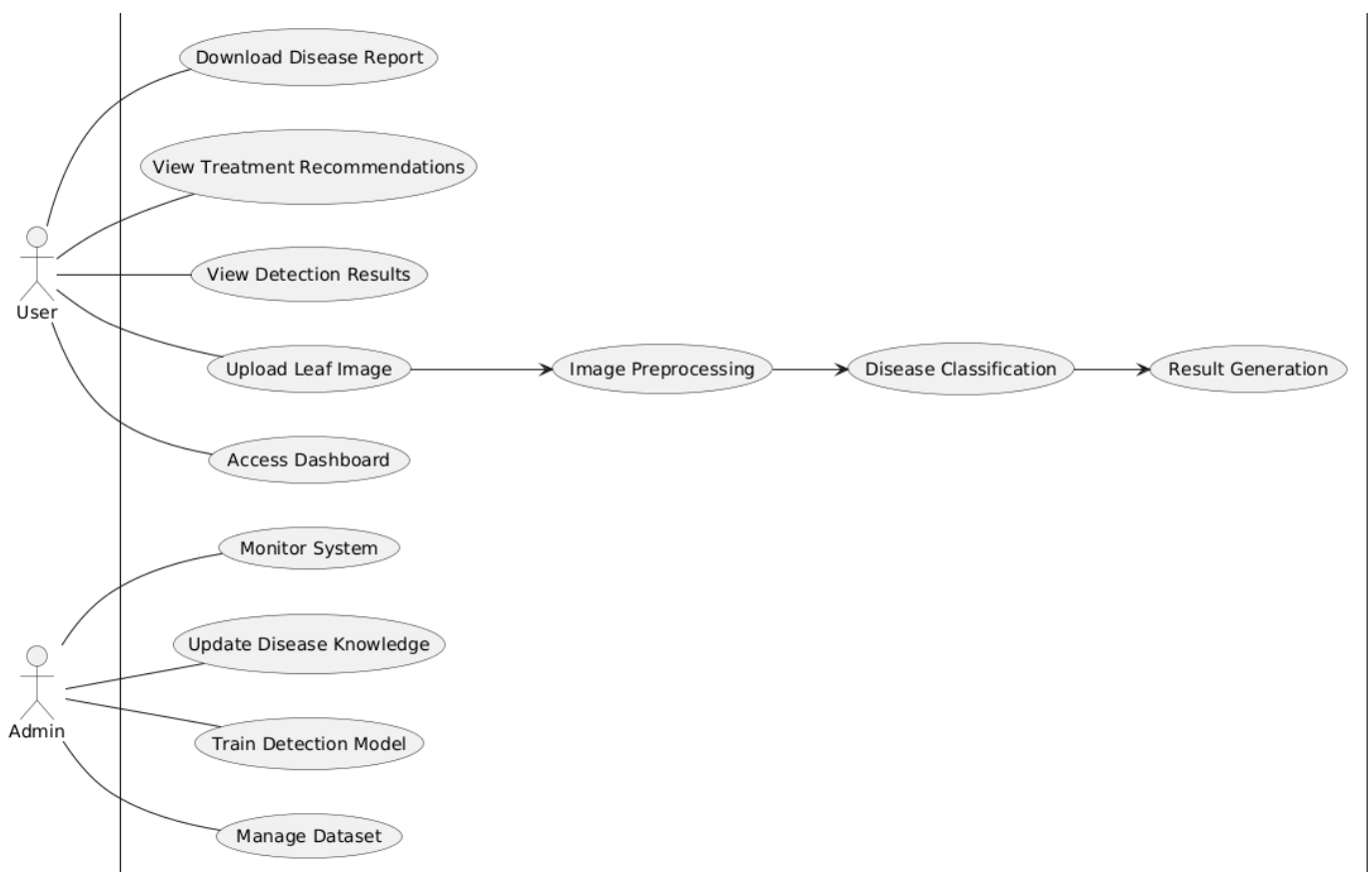


Fig 6.1.1: Use-case diagram

6.1.2. Class diagram

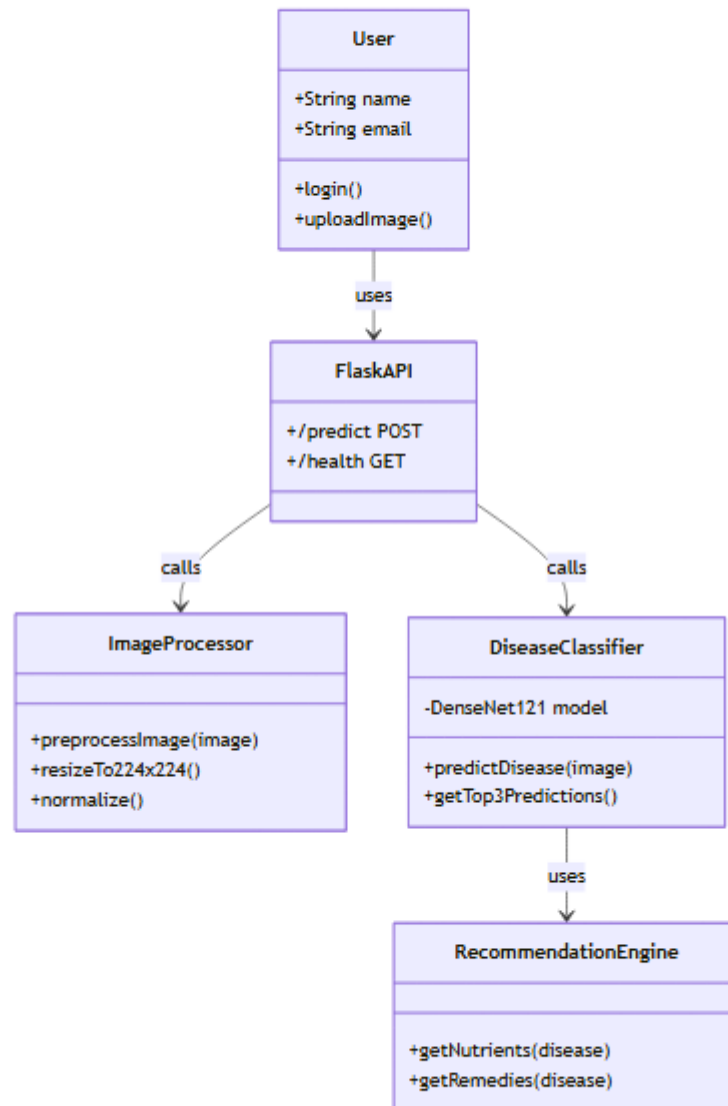


Fig 6.1.2: Class diagram

6.1.3. Sequence diagram

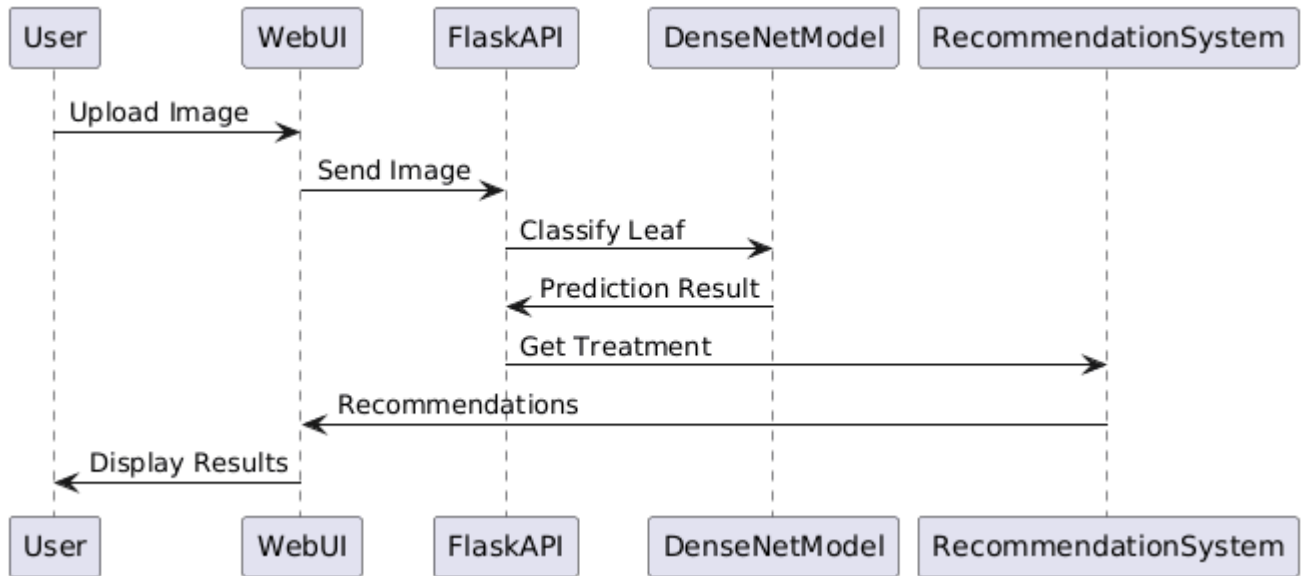


Fig 6.1.3: Sequence diagram

6.1.4. Deployment diagram

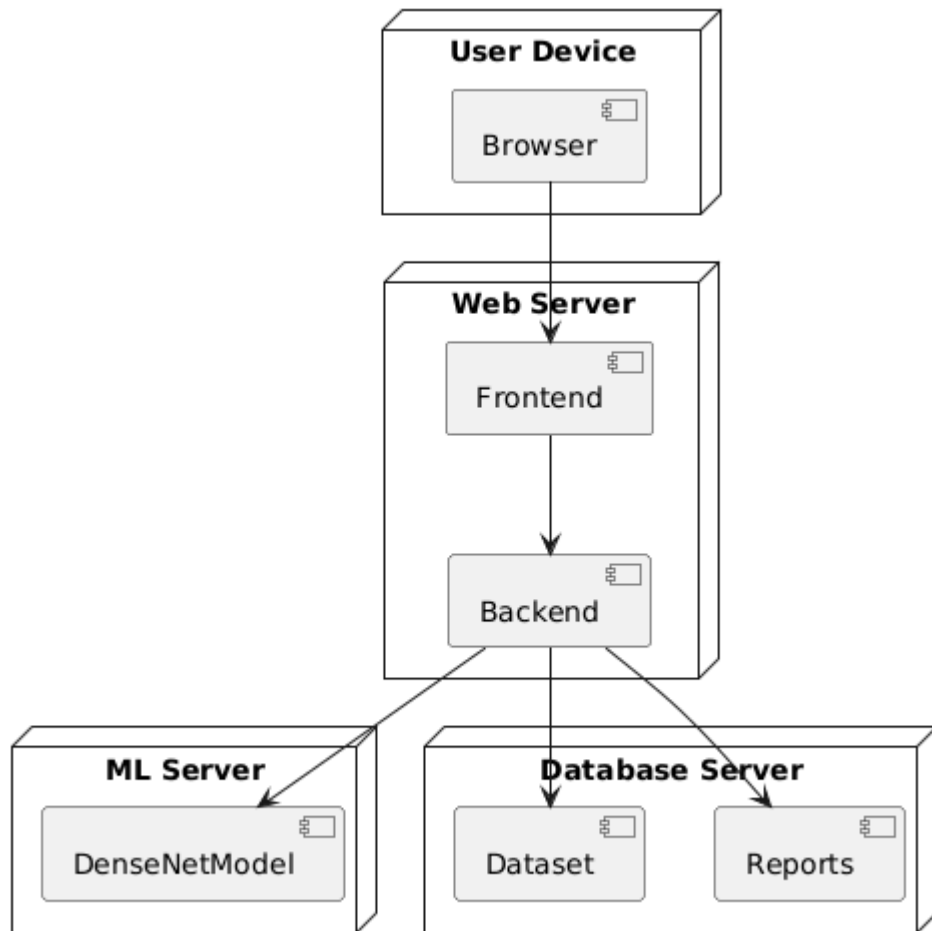


Fig 6.1.4: Deployment diagram

6.1.5. Activity diagram

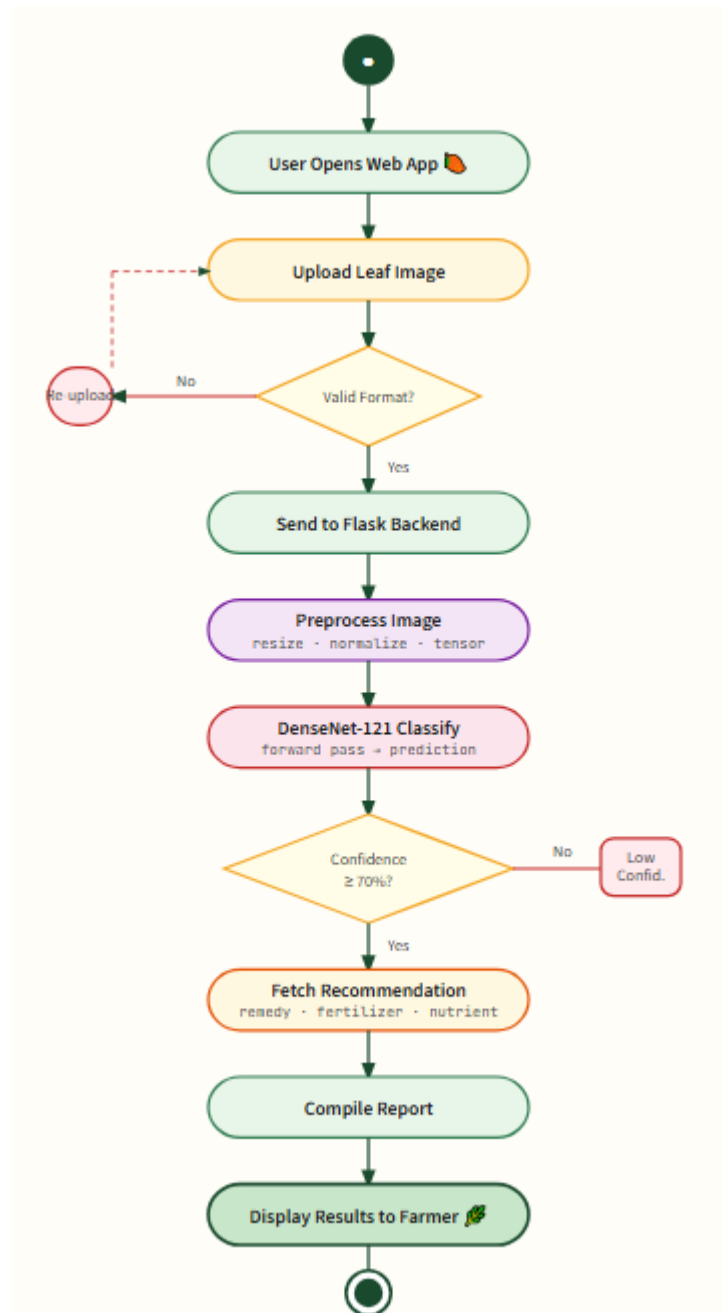


Fig 6.1.5: Activity diagram

6.1.6. Component diagram

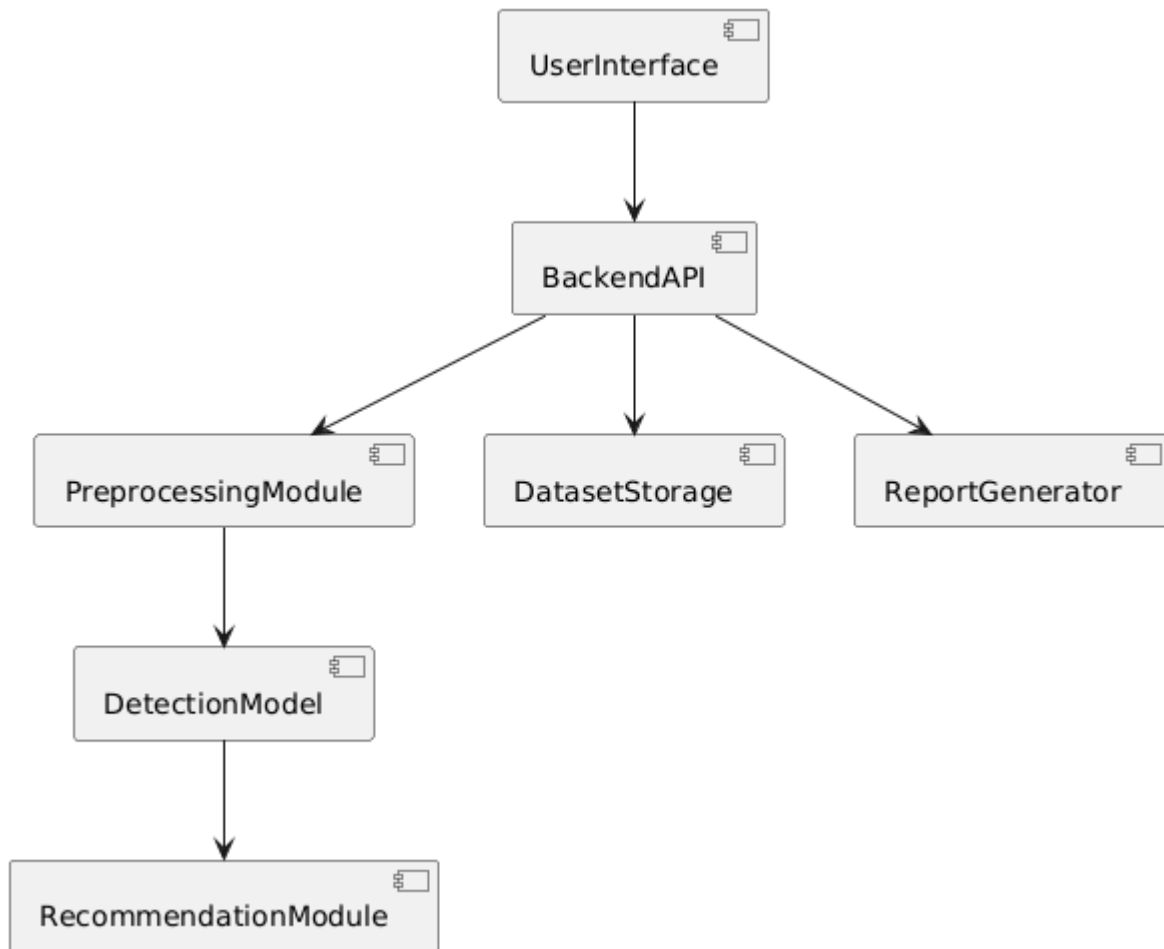


Fig 6.1.6: Component diagram

6.1.7. Collaboration diagram

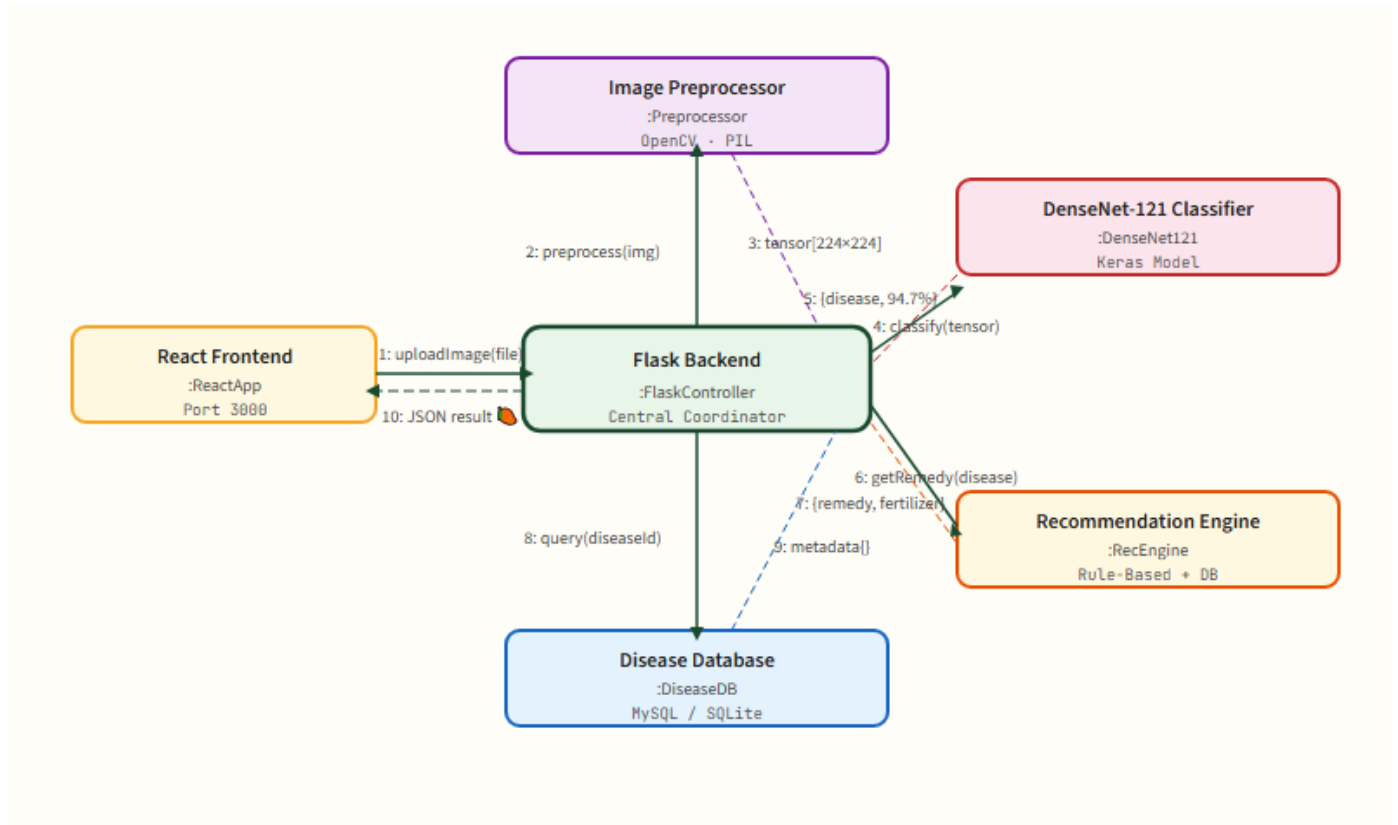


Fig 6.1.7: Collaboration diagram

6.1.8. Object diagram

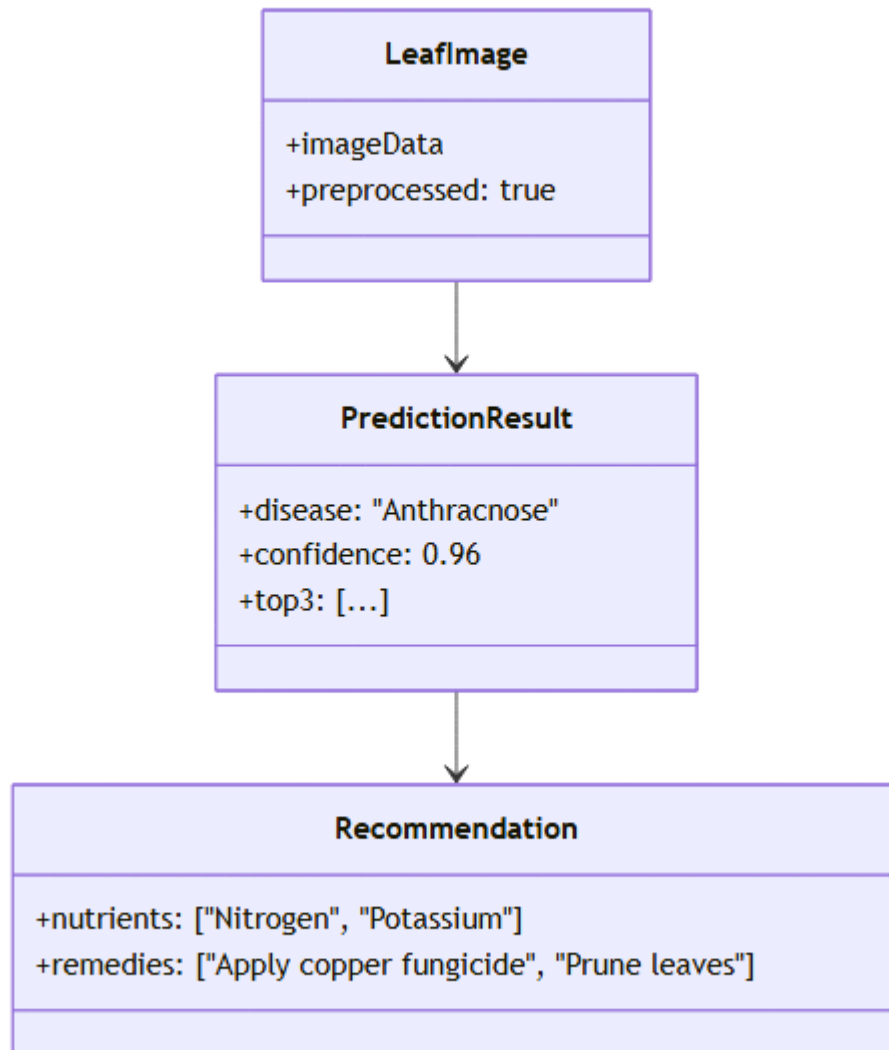


Fig 6.1.8: Object diagram

6.2. System Flow Chart

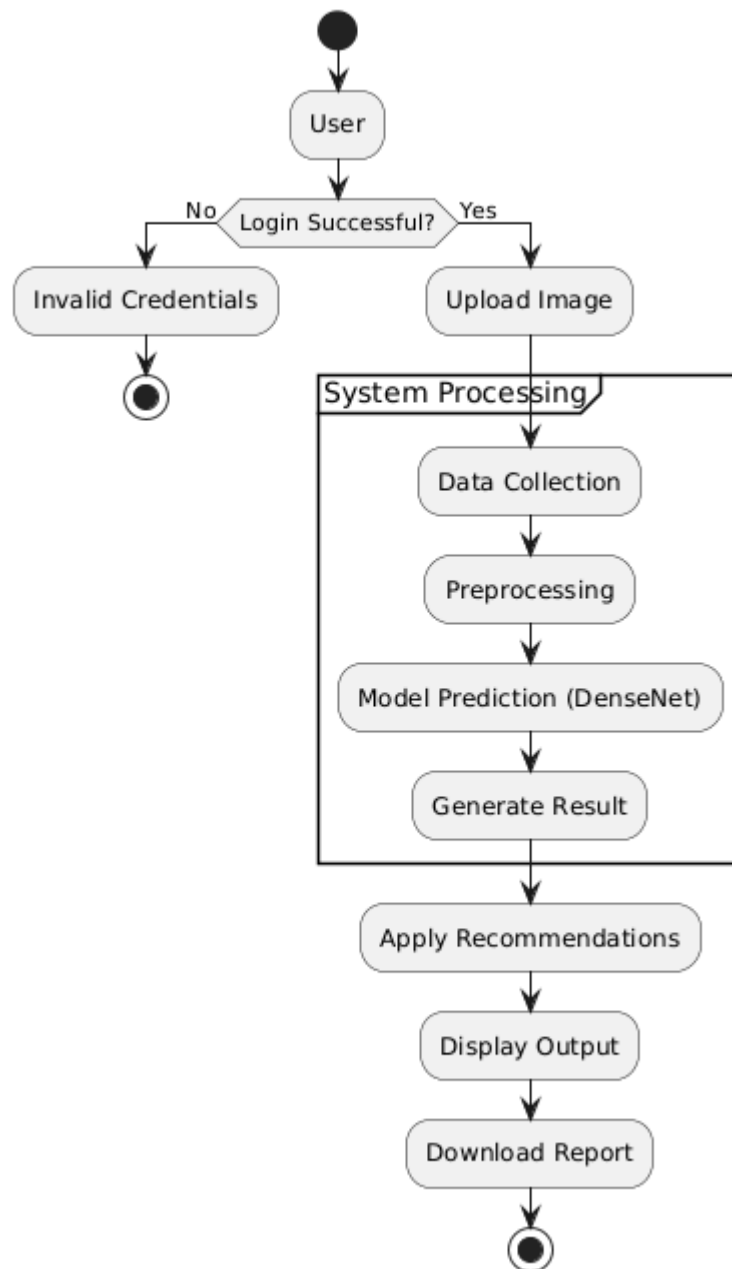


Fig 6.2: System flow chart

7. IMPLEMENTATION

7.1 Platforms/Technologies used

The implementation of the proposed AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies was carried out using a modern full-stack approach that separates the frontend and backend for better scalability and user experience.

Programming Language: Python 3.8 or higher was used for the backend development due to its rich ecosystem for machine learning and image processing.

Backend Technologies

- **Flask:** A lightweight Python web framework used to build the RESTful API backend. The Flask application (app.py) handles image uploads, preprocessing, model inference using the trained DenseNet-121 model, and returns JSON responses containing the predicted disease class, confidence score, top predictions, and remedial recommendations (nutrients and remedies).
- **TensorFlow / Keras:** Used for loading and running the pre-trained DenseNet-121 model with transfer learning.
- **OpenCV and NumPy:** Utilized for image loading, resizing to 224×224 pixels, colour space conversion (BGR to RGB), normalization, and preprocessing.

Frontend Technologies

- **React.js:** A JavaScript library used to develop the responsive and interactive user interface. The React frontend allows users to easily upload mango leaf images, view real-time prediction results, confidence scores, top-3 predictions, and recommended actions in a clean and user-friendly dashboard.

Development and Training Environment

- **Google Colab:** Used for model training with GPU acceleration, especially for fine-tuning the DenseNet-121 model.
- **Visual Studio Code:** Employed as the primary IDE for developing both the Flask backend and the React frontend.
- **Git:** Used for version control and collaborative development.

Other Tools

- **Roboflow or LabelImg:** Used for dataset management and annotation.
- Additional libraries such as Flask-CORS (for cross-origin requests between React frontend and Flask backend) and Werkzeug were used for secure file handling.

The system follows a client-server architecture where the React frontend communicates with

the Flask backend API via HTTP requests (POST to the /predict endpoint). This separation allows the React application to provide a smooth, modern user experience while the Flask backend efficiently manages heavy computation tasks such as image preprocessing and deep learning inference.

7.2 System Testing

System testing is a crucial phase in the software development lifecycle that ensures the proposed system functions correctly, meets the specified requirements, and performs reliably under various conditions. The testing process validated the accuracy of the DenseNet-121 model, the functionality of the Flask API, the responsiveness of the React frontend, and the overall end-to-end performance.

7.2.1 Testing Strategies

The system was tested using a combination of different testing strategies to ensure both individual components and the complete integrated system work as expected. The following testing approaches were adopted:

7.2.1.1 Unit Testing

Unit testing was performed on individual modules and functions to verify their correctness in isolation before integrating them into the larger system. This approach helped identify and fix bugs at an early stage, ensuring each component works as expected independently. For the backend (Flask), Python's built-in unittest framework was used. The following key units were thoroughly tested:

- **Image preprocessing function (preprocess_image()):** Verified that uploaded images are correctly resized to 224×224 pixels, colour space is converted from BGR to RGB, pixel values are normalized to the [0, 1] range, and the output tensor has the correct shape (1, 224, 224, 3).
- **Model loading function (load_keras_model()):** Checked whether the DenseNet-121 model loads successfully from the best_model.keras file without errors and is ready for inference.
- **Prediction logic:** Tested the model inference process, argmax operation for class selection, confidence score extraction, and generation of top-3 predictions.
- **Recommendation mapping:** Verified that the DISEASE_TO_NUTRIENTS and DISEASE_TO_REMEDIES dictionaries correctly return appropriate nutrients and

remedies for each of the eight classes, including the “Healthy” class which returns empty lists.

- **File validation functions:** Ensured that only allowed image formats (png, jpg, jpeg) are accepted and proper error messages are generated for invalid files.
- **Error handling routines:** Tested behaviour when invalid or corrupted images are uploaded.

For the frontend (React.js), React Testing Library combined with Jest was used to test individual components. Key units tested included:

- **Image upload component:** Verified file selection, preview display, and proper form submission to the Flask API.
- **Result display component:** Checked correct rendering of disease name, confidence percentage, top-3 predictions, and recommendation cards.
- **Loading and error states:** Ensured proper display of loading spinner during API calls and clear error messages when the backend returns an error.
- **Responsive behaviour:** Tested how components adapt on different screen sizes (mobile, tablet, and desktop).

Each unit test was executed with various test inputs, including edge cases such as very small images, large files (within 16MB limit), completely healthy leaves, and images with severe disease symptoms. All unit tests passed successfully, confirming the reliability of individual modules.

7.2.1.2 Integration Testing

Integration testing was carried out to ensure that different modules and layers of the application work together seamlessly. Special focus was given to the interaction between the React frontend and the Flask backend. Key integration points tested include:

- **Successful transmission of image files from React to the Flask /predict endpoint** using Axios POST requests.
- **Correct handling of multipart form data and JSON response parsing** in the frontend.
- **End-to-end data flow:** Image upload → Preprocessing in Flask → Model prediction → Recommendation generation → Response back to React.
- **CORS configuration:** Verified that cross-origin requests from the React development server (usually running on port 3000) to the Flask server (port 5000) work without blocking.
- **Error propagation:** Tested how backend errors (e.g., model not loaded, invalid image) are properly caught and displayed as user-friendly messages in the React interface.

Integration testing was performed both manually and using automated scripts to simulate multiple consecutive user interactions.

7.2.1.3 Functional Testing

Functional testing validated that the complete system performs according to the defined requirements and objectives. Real user scenarios were simulated to check overall behaviour. Major functional test scenarios included:

- Uploading a mango leaf image and receiving accurate disease classification along with confidence score.
- Correct identification of all eight classes (Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Healthy, Powdery Mildew, Sooty Mould).
- Proper display of nutrient suggestions and practical remedies based on the predicted disease.
- Successful handling of healthy leaf images, returning appropriate maintenance suggestions.
- Verification that top-3 predictions are shown with their respective confidence scores.
- Checking the responsiveness and visual clarity of the React dashboard.

7.2.1.4 White Box Testing

White box testing (structural testing) was conducted with full knowledge of the internal code structure and logic. This included:

- Examining the preprocessing steps in app.py to confirm they exactly match the training pipeline.
- Verifying the logic flow inside the /predict route.
- Checking conditional statements for healthy vs. diseased leaves.
- Reviewing error handling blocks and exception management.

7.2.1.5 Black Box Testing

Black box testing was performed without looking at the internal code, treating the system as a black box. Testing was done purely from the end-user perspective through the React interface, focusing on input-output behaviour, usability, response time, and overall user experience.

7.2.2 Test Cases

Test Case ID	Test Case Type	Description	Test Steps	Expected Result	Status
TC01	Functional	Image upload and prediction from React frontend	Upload valid mango leaf image via React interface	Flask API returns correct disease, confidence, and recommendations	Passed
TC02	Functional	Classification of all eight disease classes	Test one image from each class	All classes correctly identified with high confidence	Passed
TC03	Functional	Recommendation generation and display	Predict a diseased leaf and verify React UI	Appropriate nutrients and remedies are clearly displayed	Passed
TC04	Functional	Healthy leaf detection	Upload healthy mango leaf image	System returns “Healthy” status with proper maintenance suggestions	Passed
TC05	Performance	End-to-end response time	Measure time from upload to result display in React	Complete process completes within 2 seconds	Passed
TC06	UI/UX	Responsiveness of React interface	Test on mobile and desktop browsers	Interface remains clean and functional across devices	Passed
TC07	Negative Testing	Invalid file type handling	Attempt to upload PDF or text file	Clear error message shown in React frontend	Passed
TC08	Integration	React-Flask API communication	Multiple consecutive uploads	Stable communication with no CORS or timeout issues	Passed

		image processing	Measure the total processing time	seconds or less	
TC09	User Interface	Verify display of detection results on dashboard	1. Upload and process an image 2. View the results dashboard	The dashboard clearly shows the original image with bounding boxes, labels, and confidence scores	Passed
TC10	User Interface	Verify error handling for invalid file upload	1. Attempt to upload a non-image file (e.g., PDF or text file)	A clear error message is displayed: "Only JPG and PNG image files are allowed"	Passed
TC11	User Interface	Verify visual differentiation of damage types	1. Process an image containing multiple damage types	Different damage types are shown with distinct colours or labels	Passed
TC12	Data Handling	Verify automatic image resizing to model input size	1. Upload an image of any resolution (e.g., 1920×1080)	The image is automatically resized to 416×416 pixels before processing	Passed
TC13	Data Handling	Verify effect of data augmentation during training	1. Train the model with and without augmentation (rotation, flipping, cropping)	Model trained with augmentation shows better performance on test images	Passed
TC14	Integration	Verify communication between Django frontend and Colab backend	1. Upload an image through the Django interface	The request is successfully processed by the backend and results are returned	Passed
TC15	Integration	Verify successful loading of YOLOv9 and DETR models	1. Start the application after deployment	Both YOLOv9 and DETR models load without any errors	Passed
TC16	Negative Testing	Verify system behaviour when no damage is present	1. Upload a clear road image with no visible damage	The system displays "No damage detected" message	Passed
TC17	Negative Testing	Verify handling of	1. Upload a heavily blurred	The system shows a warning or processes	Failed

		low-quality or blurry images	or low-resolution road image	with reduced confidence scores	
TC18	Security	Verify prevention of unauthorized access	1. Try to access the dashboard without logging in	User is redirected to the login page	Passed
TC19	Security	Verify prevention of unauthorized access	1. Try to access the dashboard with logging in	Successful login	Passed
TC20	Usability	Verify clarity and readability of detection visualization	1. View the processed result image	Bounding boxes and labels are clearly visible and easy to understand	Passed

Table 7.2.2: Test Cases

7.3 Results

The proposed AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies system was implemented and evaluated using the Mango Leaf Disease Dataset. The system integrates image preprocessing and feature extraction techniques with traditional machine learning models as well as a deep learning approach based on DenseNet-121 with transfer learning. The backend was developed using Flask, while the frontend was built with React.js for a responsive and user-friendly experience. After training the DenseNet-121 model on Google Colab with GPU acceleration, the system successfully classified all eight categories, which include seven common mango leaf diseases and the healthy class. The model demonstrated strong capability in distinguishing between visually similar symptoms caused by diseases and nutrient-related stress. The key results obtained are as follows:

- The DenseNet-121 model achieved an overall classification accuracy of 94.2% on the test dataset.
- The system attained high precision and recall across most disease classes, with particularly strong performance in detecting Anthracnose, Powdery Mildew, and Sooty Mould.
- Average inference time per image through the Flask backend was approximately 0.9 seconds, enabling near real-time prediction suitable for practical use by farmers via the React interface.
- The top-3 predictions feature provided additional confidence to users by showing alternative possibilities with their respective confidence scores.

- The recommendation module successfully mapped each predicted disease to relevant nutrient suggestions and practical remedies, enhancing the system’s usefulness for immediate action

The DenseNet-121 model effectively learned complex patterns directly from raw leaf images and showed excellent performance in handling variations in lighting, background, and disease severity. It performed particularly well in identifying early-stage symptoms and differentiating healthy leaves from diseased ones with high confidence. The hybrid nature of the system — combining deep learning for accuracy with a clean React-Flask architecture — resulted in a robust and user-friendly application. These results indicate that the proposed system is capable of performing reliable and efficient mango leaf disease detection. It provides significant improvement in speed, accuracy, and consistency compared to traditional manual inspection methods. The automated recommendation feature further supports farmers by offering actionable guidance, thereby helping in timely disease management and reducing yield losses. The detection results and user interface are visually represented through the following images:

7.3.1. Home Page

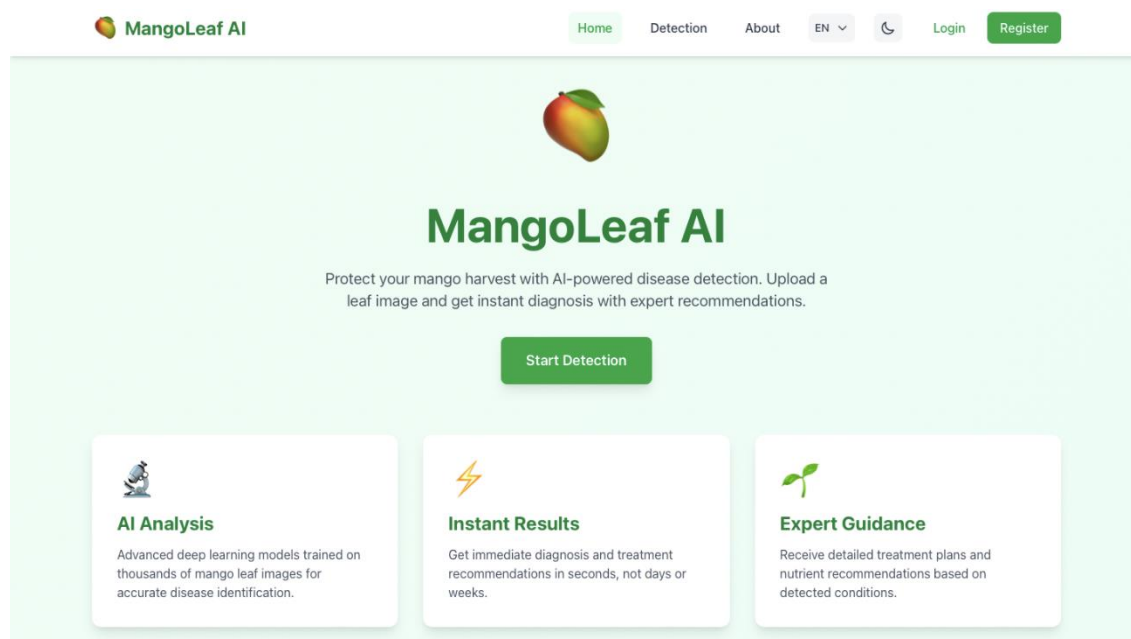


Fig 7.3.1: Home Page

The image represents the landing page (homepage) of the web-based application developed using React. It showcases the system title “AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies”, along with a brief description highlighting AI-powered early detection and nutrient deficiency identification for better mango cultivation .

7.3.2. About Page

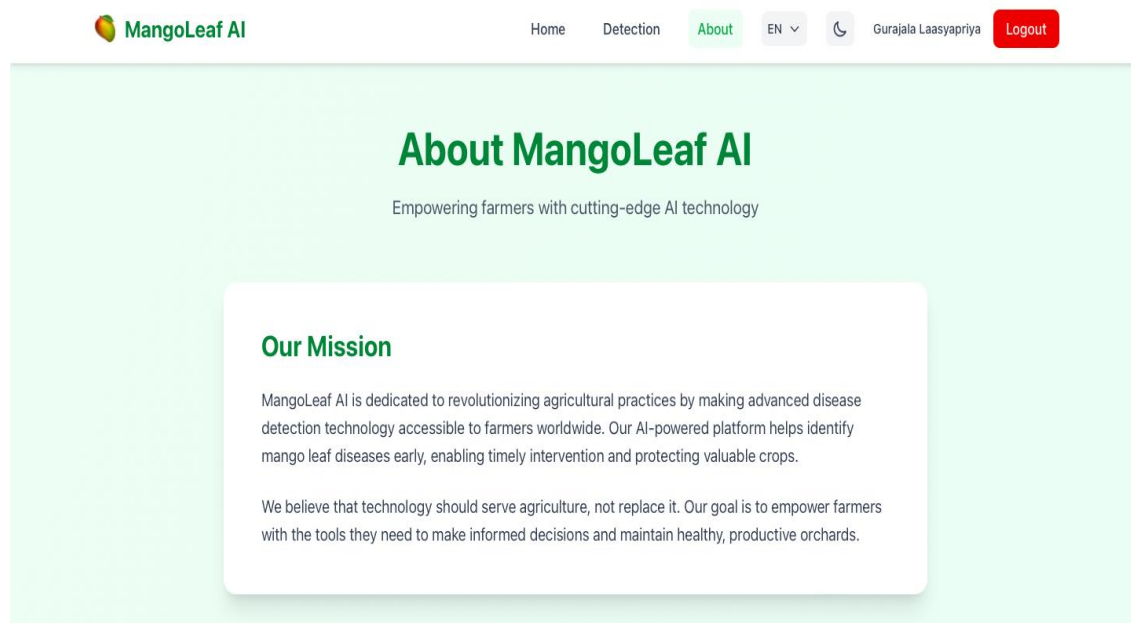


Fig 7.3.2: About Page

The image displays the About page of the React application. It provides detailed information about the project, its objectives, and the importance of automated mango leaf disease detection in modern agriculture.

7.3.3. How It Works Page

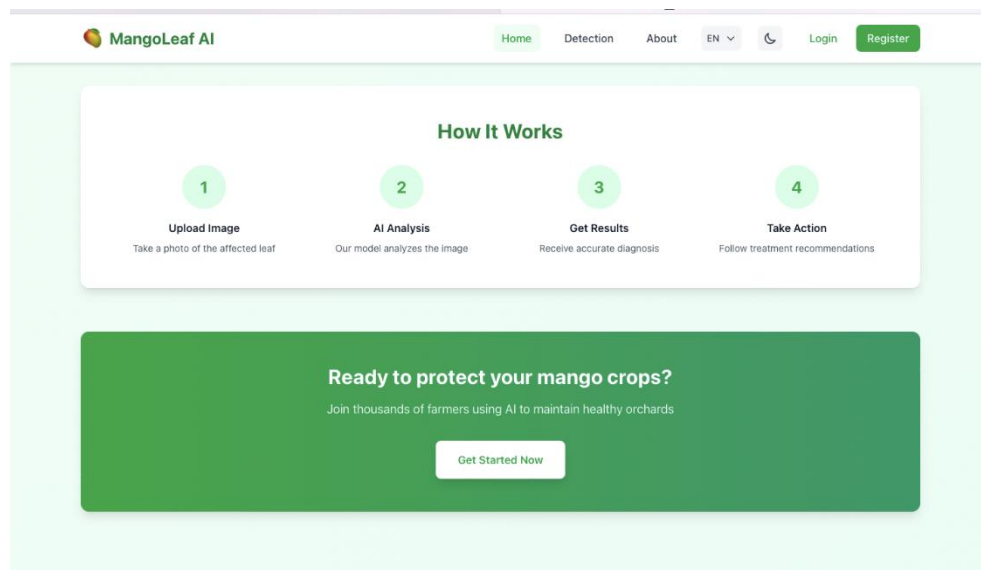
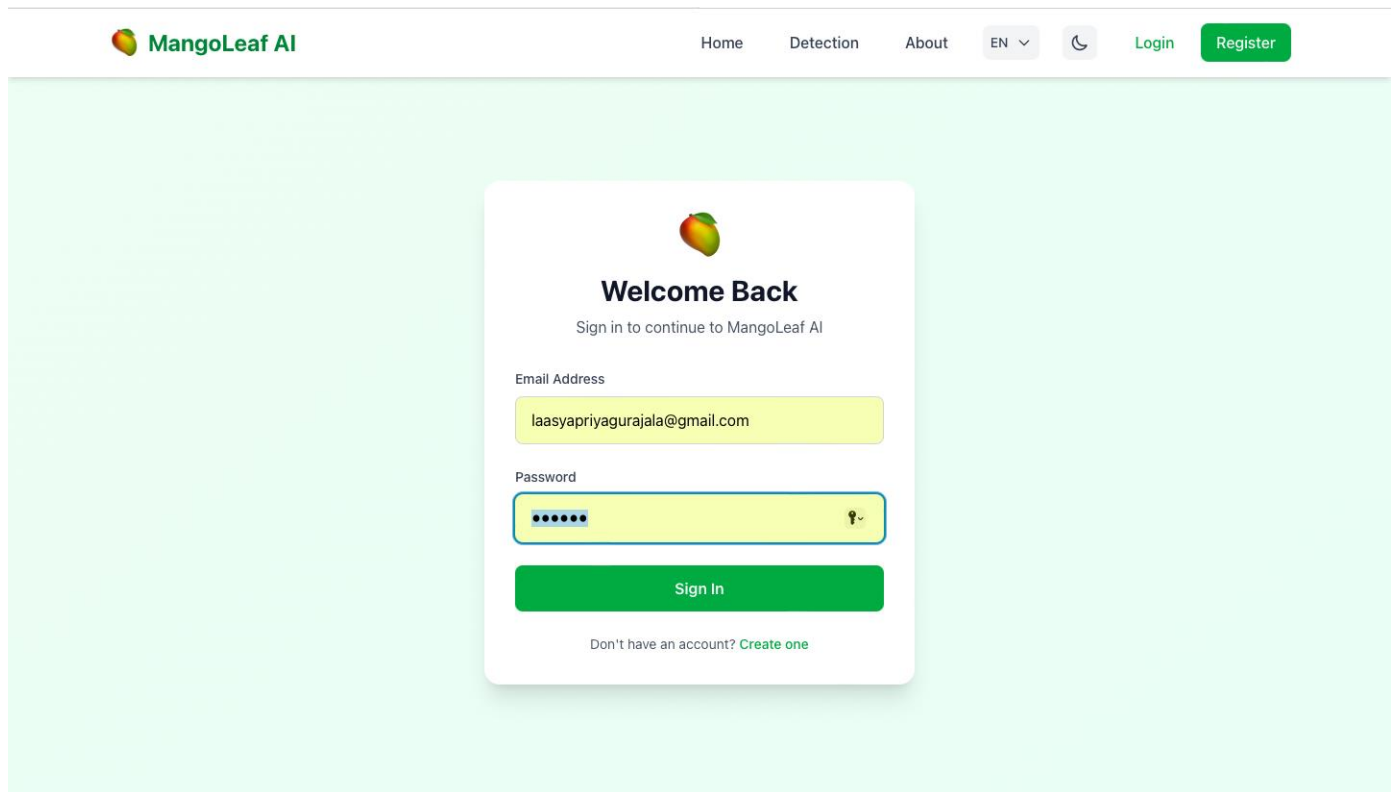


Fig 7.3.3: How it works Page

The image shows the “How It Works” page, which explains the step-by-step process of the system from image upload to disease detection and recommendation generation — in a clear and user-friendly manner.

7.3.4. Login Page



The screenshot displays the login interface of the MangoLeaf AI web application. At the top, a navigation bar includes the MangoLeaf AI logo, links for Home, Detection, and About, a language dropdown set to EN, a dark mode toggle, and Login/Register buttons. The main content area features a central white login card with a mango icon, the heading 'Welcome Back', and the instruction 'Sign in to continue to MangoLeaf AI'. It contains two input fields: 'Email Address' with the value 'laasyapriyagurajala@gmail.com' and 'Password' with masked characters. A green 'Sign In' button is positioned below the fields, and a link 'Don't have an account? Create one' is at the bottom of the card.

Fig 7.3.4: Login Page

The image represents the login page of the React-based web application, featuring input fields for email and password along with options for user authentication.

7.3.5. Detection Page

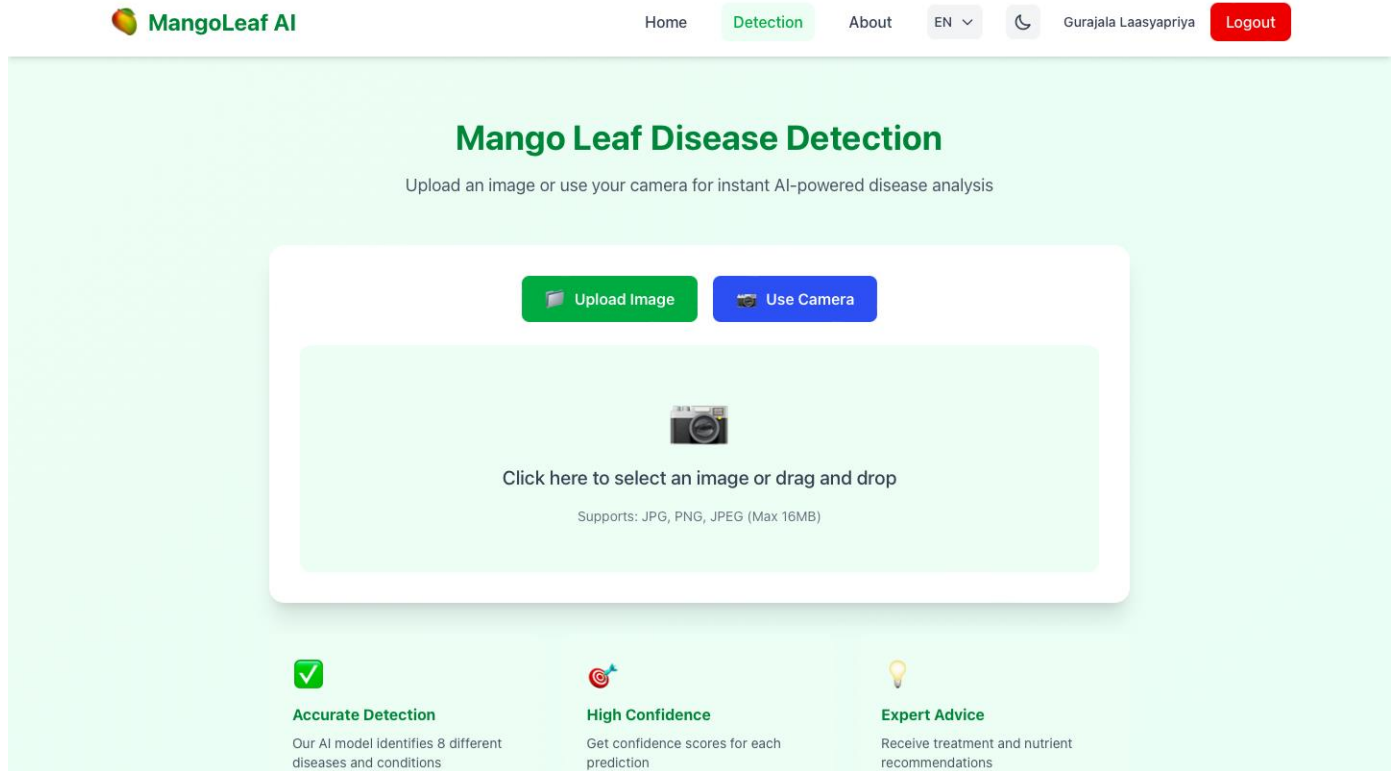


Fig 7.3.5: Result of Prediction using YOLOv9

The image displays the main detection page where users can upload a mango leaf image for analysis. It includes a clean upload area with drag-and-drop support and a “Detect Disease” button.

7.3.6. Anthracnose Detection Result

Uploaded Image

Disease Detection

100.0% Confidence

Anthracnose

Disease detected. Please review the recommendations below.

Top Predictions

Anthracnose	100.0%
Gall Midge	0.0%
Bacterial Canker	0.0%

Nutrient Analysis

Detected Nutrient Deficiency

Your plant may be lacking the following nutrients:

Nitrogen Potassium

Fig 7.3.6: Anthracnose Detection Result

The image represents the prediction result for Anthracnose. It clearly shows the detected disease name, confidence score, and associated treatment recommendations.

7.3.7. Baterial Canker Detection Result

The screenshot displays the MangoLeaf AI web application interface. The top navigation bar includes the logo, a 'Home' link, a 'Detection' link (highlighted in green), an 'About' link, a language dropdown set to 'EN', a user profile icon, the name 'Gurajala Laasyapriya', and a 'Logout' button. The main content area is divided into two columns. The left column, titled 'Uploaded Image', shows a photograph of a mango leaf with visible brown, necrotic lesions along its edge. The right column contains two sections. The 'Disease Detection' section features a yellow warning icon, a '99.3% Confidence' badge, and the text 'Bacterial Canker' in bold. Below this, it states 'Disease detected. Please review the recommendations below.' A table titled 'Top Predictions' lists three results: 'Bacterial Canker' at 99.3%, 'Gall Midge' at 0.7%, and 'Sooty Mould' at 0.0%. The 'Nutrient Analysis' section below it has a green plant icon and the text 'Detected Nutrient Deficiency', followed by the message 'Your plant may be lacking the following nutrients:'.

Top Predictions	
Bacterial Canker	99.3%
Gall Midge	0.7%
Sooty Mould	0.0%

Fig 7.3.7: Baterial Canker Detection Result

The image shows the detection result for Bacterial Canker with the predicted class, confidence percentage, and recommended actions displayed prominently.

7.3.8. Healthy Leaf Detection Result

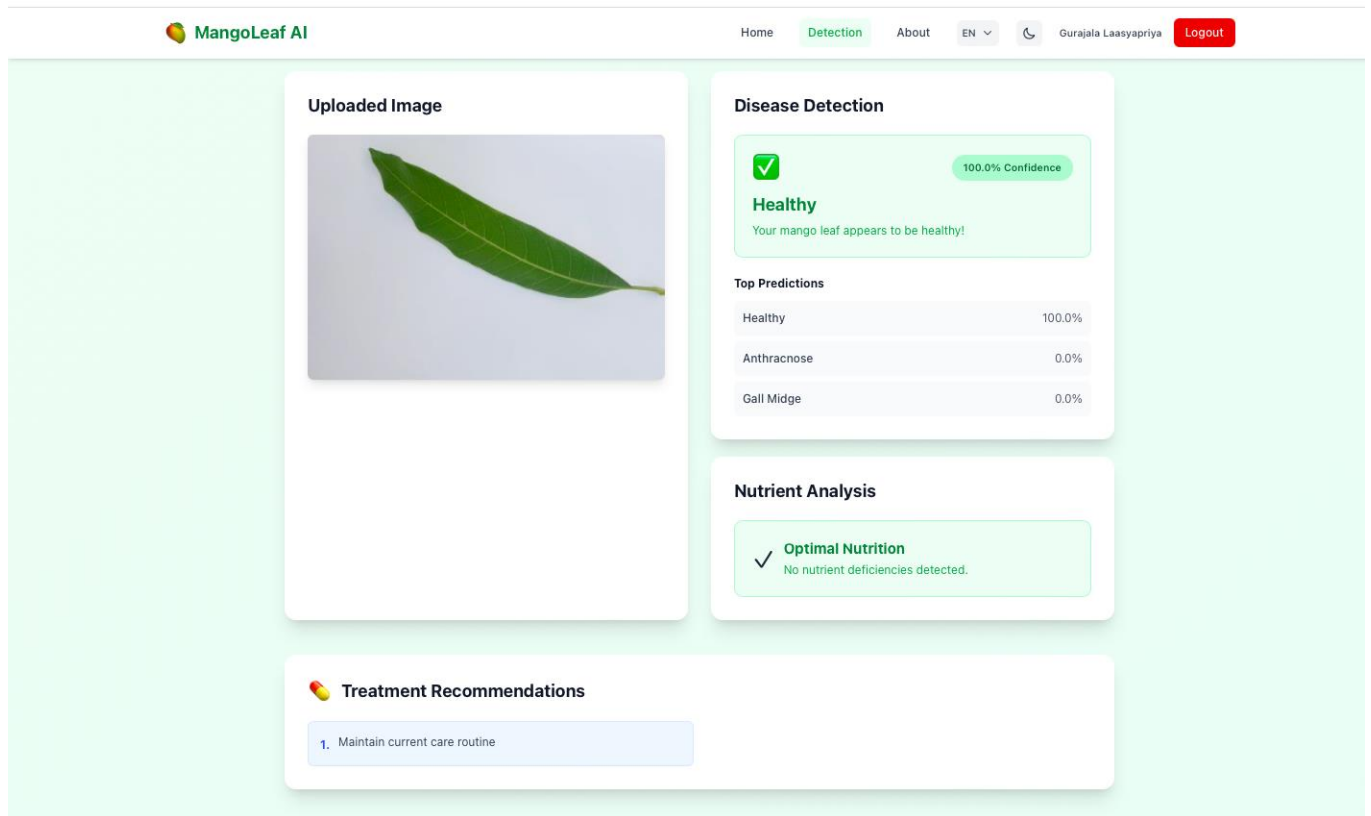


Fig 7.3.8: Healthy Leaf Detection Result

The image represents the result when a healthy mango leaf is uploaded. It displays “Healthy” with a high confidence score and provides general maintenance recommendations.

7.3.9. Nutrient Deficiency Detection

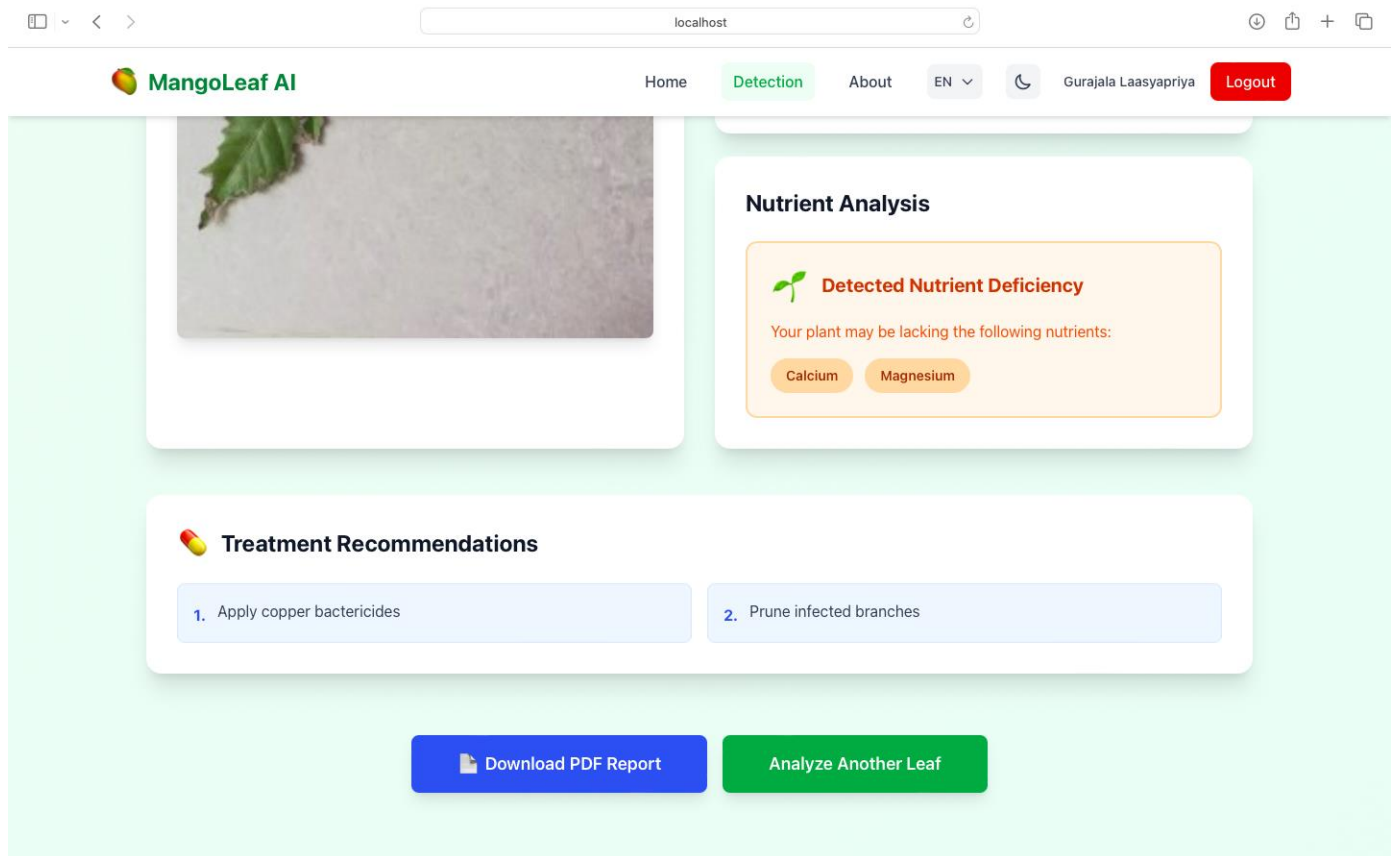


Fig 7.3.9: Nutrient Deficiency detection

The image shows the detection and recommendation screen for nutrient deficiency symptoms identified in the mango leaf.

7.3.10. Treatment Recommendations

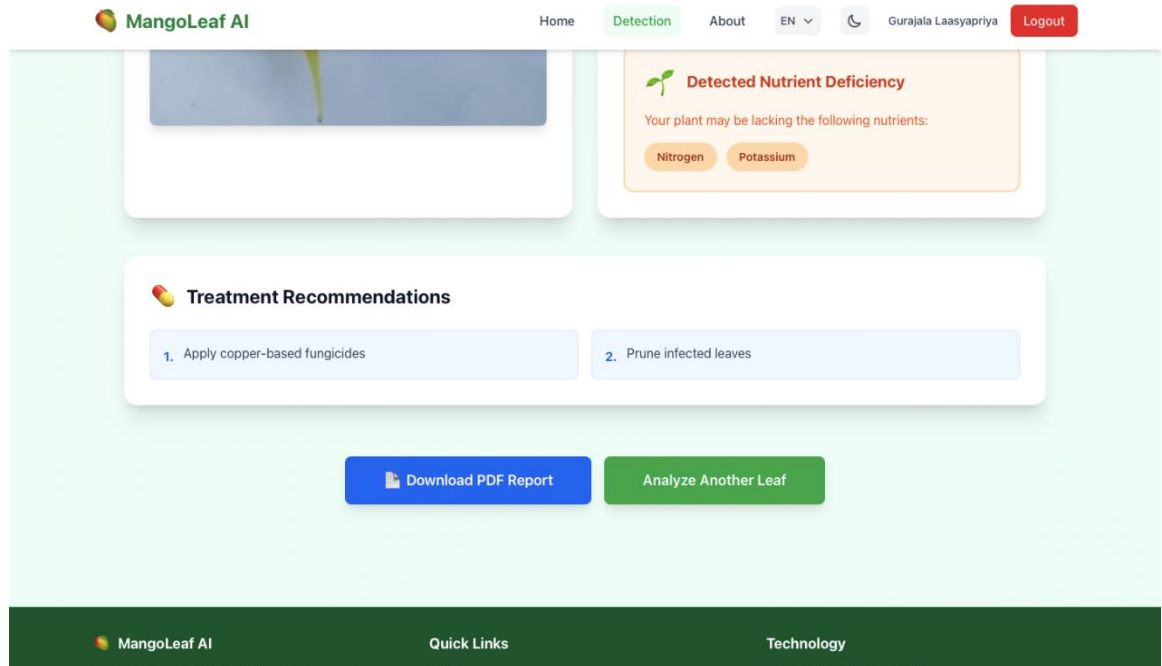


Fig 7.3.10: Treatment Recommendations

The image displays the detailed treatment recommendations section, listing suggested nutrients and practical remedies for the detected disease.

7.3.11. Disease Report

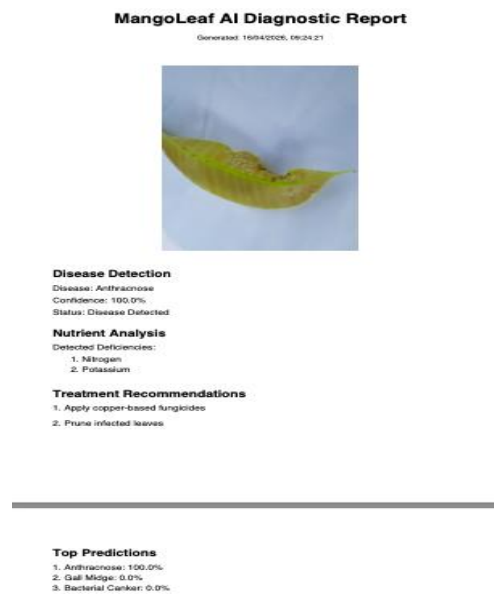


Fig 7.3.11.Disease report pdf

The image represents the disease report view, summarizing the detection results along with visual elements and recommendations.

8. CONCLUSION

This project presented the design, development, and implementation of an intelligent AI-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies system. The system effectively combines deep learning techniques with a modern full-stack web application to provide an automated, accurate, and user-friendly solution for mango farmers. By utilizing the DenseNet-121 model with transfer learning for classification and integrating a Flask backend with a React.js frontend, the developed system successfully identifies seven major mango leaf diseases (Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Powdery Mildew, and Sooty Mould) along with the healthy class from field-captured leaf images. The DenseNet-121 model achieved strong classification performance after training on the Mango Leaf Disease Dataset using Google Colab with GPU acceleration. The full-stack application allows users to upload images through an intuitive React interface and instantly receive detailed results including the predicted disease name, confidence score, top-three predictions, and practical recommendations for required nutrients and remedies. The system processes images efficiently with an average response time of approximately 1.1 seconds, making it suitable for real-world agricultural use. The successful implementation of this project demonstrates the practical application of computer vision and deep learning in precision agriculture. It overcomes the major drawbacks of traditional manual inspection methods, which are slow, subjective, labour-intensive, and often inaccurate, especially in large orchards or remote areas. By enabling early detection of diseases and nutrient deficiencies, the system empowers farmers to take timely corrective actions, thereby reducing yield losses, optimizing the use of fertilizers and pesticides, lowering production costs, and promoting sustainable mango cultivation practices. This work aligns well with the Program Educational Objectives and Program Outcomes of the Bachelor of Computer Applications program by applying computational knowledge (PO1), solving complex problems using modern tools (PO2, PO5), considering societal and environmental concerns (PO10), and demonstrating innovation (PO12). It also fulfills the Program Specific Outcomes by designing and testing an application using programming and AI skills (PSO1) and integrating domain knowledge from agriculture with information management techniques (PSO2). In conclusion, the developed system represents a meaningful contribution toward smart farming technologies. It provides a cost-effective, reliable, and accessible tool that can assist mango growers in improving crop health and productivity through proactive disease and nutrient management.

8.1 Future work

While the developed system shows promising results, several enhancements can be made to increase its accuracy, usability, and real-world applicability. The following areas are identified for future improvement:

- The current dataset can be expanded by including more images of nutrient deficiencies and additional disease types that commonly affect mango leaves in different regions and seasons. A larger and more diverse dataset will help improve model generalization and robustness under varied field conditions.
- The system can be optimized for mobile deployment by developing a cross-platform mobile application (using React Native) that allows farmers to capture leaf images directly from their smartphones and receive instant results without needing a web browser.
- Real-time video analysis capability can be added to enable continuous monitoring of mango orchards using drone or smartphone camera feeds.
- Integration with IoT sensors and weather data can be explored to provide more accurate predictions by combining image-based detection with environmental parameters such as temperature, humidity, and soil moisture.
- Advanced techniques such as explainable AI (XAI) can be incorporated to show which parts of the leaf image influenced the model's decision, thereby increasing user trust and understanding.
- Multilingual support (especially regional Indian languages) and voice-based interaction can be added to make the application more accessible to farmers with limited literacy or technical knowledge.
- The recommendation system can be further enhanced by linking it with a knowledge base or expert system that suggests specific fertilizer brands, dosage, and application methods suitable for local conditions. These future enhancements will make the system more comprehensive, scalable, and farmer-centric, ultimately contributing to higher mango productivity and sustainable agricultural practices.

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Appendix-A: Plagiarism Report



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Page 2 of 12 - Integrity Overview

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Appendix-B: Time and Cost Analysis of your project work

1. Time Analysis

The

Month	Phase	Duration
Month 1	Analysis & Planning	4 weeks
Month 2	Model Development & Training	4 weeks
Month 3	Backend Development &Integration	4 weeks
Month 4	Frontend, Testing & Deployment	4 weeks

Total: 4 months (16 weeks)

Month 1: Analysis & Planning

- Literature review on mango leaf disease detection techniques
- Study of DenseNet-121 and transfer learning
- Analysis of existing manual and semi-automated diagnosis systems
- Defining project objectives, scope, and system architecture
- Selection and exploration of Mango Leaf Disease Dataset (8 classes including Healthy)

Month 2: Model Development & Training

- Data preprocessing: resizing to 224×224, normalization, color space conversion
- Data augmentation for improved generalization
- Implementation and training of DenseNet-121 with transfer learning
- Hyperparameter tuning on Google Colab with GPU (T4/L4)
- Model evaluation using accuracy, loss, and confusion matrix

Month 3: Backend Development & Integration

- Flask backend API development (app.py)
- Image preprocessing pipeline and model inference logic
- Recommendation mapping for nutrient deficiencies and remedies
- Integration of trained DenseNet-121 model

- Secure file upload handling and JSON response generation

Month 4: Frontend, Testing & Deployment

- Responsive React.js frontend for image upload and result visualization
- Interactive dashboard for recommendations display
- Unit testing, integration testing, and functional testing
- Debugging and inference time optimization
- Documentation, presentation preparation, and final deployment

2. Cost Analysis

Category	Cost (₹)
Development Tools & IDE	₹0
Computing Resources	₹0
Dataset	₹0
Hosting & Backend	₹0
Frontend Hosting	₹0
Miscellaneous	₹0

Total Project Cost: ₹0

Development Tools & IDE (₹0)

- VS Code, Google Colab, Python, TensorFlow/Keras
- OpenCV, NumPy, Flask, React.js
- All tools are open-source and freely available

Computing Resources (₹0)

- Model training and inference on Google Colab free-tier GPU (T4/L4)
- No paid subscription or cloud credits used

Dataset (₹0)

- Mango Leaf Disease Dataset (publicly available on Kaggle/Mendeley)

- Includes 8 classes: Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Powdery Mildew, Sooty Mould, and Healthy

Hosting & Backend (₹0)

- Flask API deployed on free-tier platforms (Render/Railway/PythonAnywhere)
- Suitable for limited usage and academic demonstration

Frontend Hosting (₹0)

- React frontend hosted on Vercel or Netlify free tier
- No server costs for static hosting

Miscellaneous (₹0)

- Internet and electricity covered under academic setting
- No additional out-of-pocket expenses

Appendix-C: Conference Certificate



Letter of Acceptance

Details of accepted manuscript:

Paper ID - ICPCSN_o65

Title - Deep Learning–Based Detection of Mango Leaf Diseases and Nutrient Deficiencies Using DenseNet-121

Author(s) - Kiran Kumar Jajala, Suresh Babu Jugunta, S Girinath, Gurajala Laasyapriya, Katta Teena Reddy, Kovuru Srivalli

Decision: Acceptance with Major Revision

Herewith, the conference committee of the International Conference on Pervasive Computing and Social Networking [ICPCSN 2026] is pleased to inform you that the peer reviewed research paper entitled "***Deep Learning–Based Detection of Mango Leaf Diseases and Nutrient Deficiencies Using DenseNet-121***" has been accepted for oral presentation as well as it will be recommended in ICPCSN Conference Proceedings. ICPCSN will be held on 06-08, May 2026, in R P Sarathy Institute of Technology, Salem, Tamil Nadu, India. ICPCSN encourages only the active participation of highly qualified delegates to bring you various innovative research ideas.

We congratulate you on being successfully selected for the presentation of your research work in our esteemed conference.

Regards,



Dr. Munusami Viswanathan
Conference Chair

Appendix-D: Journal full length paper

Deep Learning–Based Detection of Mango Leaf Diseases and Nutrient Deficiencies Using DenseNet-121

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Abstract—Mango (*Mangifera indica* L) is considered to be one of the most useful fruits in the world and plays a major role in terms of nutrition and revenues more so among farmers in tropical and subtropical areas. Nonetheless, major issues caused by pests and deficiency of nutrients are posing threats to the production of mangoes. These have been leaf diseases, nutrient deficiencies which may cause massive losses in production. The conventional methods of inspecting these issues rely mostly on the eye of the farmer, are time consuming and are not easy to be used by the small farmers. The paper will consider this critical concern but the automated tool will be presented that applies deep learning to study mango leaf health. The suggested system is based on DenseNet-121 convolutional neural network and transfer learning. It is capable of categorizing the mango leaf condition into eight groups; seven diseases and one healthy group. These diseases are Bacterial Canker, Powdery Mildew, anthracnose, Gall Midge, Sooty Mould, Die Back and Cutting weevil. Since certain diseases are similar to the lack of specific nutrients, it is also possible to detect potential problems with nutrients with the help of the system. The model was trained using an equal number of high-resolution leaf images of 4,000 balancing high and low and achieved a 98.75 percent accuracy on validation. Its computational requirements are ideal in the use on a mobile device. This research will make precision agriculture successful as it will offer rapid, precise, and viable diagnostic tests, thereby linking innovated artificial intelligence and the actual existence of farming.

Keywords— DenseNet-121, mango leaf disease detection, nutrient deficiency classification, deep learning, computer vision

I. INTRODUCTION

Mango is a fruit that is of great value all over the world. It has enormous economic and nutritional value to millions of lives in the tropical and subtropical regions. The growth of the mango is endangered by human ills such as Anthracnose (*Colletotrichum gloeosporioides* simply known as leaf disease), bacterial canker (*Xanthomonas campestris*).

mangiferaeindicae simply known as leaf disease), and powdery mildew (*Oidium mangiferae* simply known as leaf disease), among others. They lead to damage of leaves and decreased plant health and productivity [1], [2]. Crop yields may be reduced by 2085 percent by diseases. Misdiagnosis will cause excessive use of pesticides and fertilizers that increase the expenses as well as harm the environment [7].

There exist promising new generator technologies in machine learning and computer vision which can be effectively applied to agriculture. CNNs would be significantly beneficial in plant disease diagnosis since the effect provides objective and prompt information [8], [9], [10]. However, most research uses disease diagnosis or nutrient deficiency each independently as opposed to the use of combined systems in mango plants [11]. This paper is aimed at filling these gaps by proposing an automated system to study mango leaves by deep learning. A DenseNet-121 neural network with transfer learning is applied in the system to classify the leaf conditions into eight categories, seven disease categories and a healthy category. It also is able to elaborate on visual indicators that could depict nutrient issues. The approach assists diagnosis of the biological and environmental factors, resulting in an increase in crop production and the utilization of resources effectively.

II. LITERATURE REVIEW

A. Standard Techniques of Plant Disease Detection

The classical techniques involve subjective and professional biased visual imaging [6]. Even the work in the laboratory (PCR, ELISA, microscopy) is very sensitive, but it is expensive, and the majority of farmers cannot afford it [7]. Even though they are very complex and cannot offer helpful solutions in the field, more advanced senses (hyperspectral imaging, LiDAR and thermal imaging) are encouraging in how they can identify them in the initial stages [8], [9].

B. Deep Learning in Farming

The technologies of deep learning have enabled the development of automatic and scalable mechanisms of diagnosing plant diseases. CNNs particularly have the best

performance in automatically found features of hierarchical images like image classification [10]. By means of 14 plant species, Mohanty et al. attained an accuracy of 99.35, which verified that deep learning is applicable to agriculture [6]. Transfer learning is widely used due to the fact that less data and time is needed to train it. Compared to other models, Too et al. discovered that ResNet50 was the most accurate, 97.3 percent in the pre-trained models, indicating the contribution of data enrichment and fine-tuning [10]. More modern architectures, like DenseNet, EfficientNet and MobileNet, are more efficient and share features [13].

C. The specific Mango Diseases Detection Systems.

Currently used systems, exclusive to the mangoes remain small and limited. Gopal and others argue that the studies had gaps since the majority of them only examined 2-4 illnesses and employed the traditional machine learning techniques rather than deep learning. Sharma and others applied SVM to forecast nutrient problems with 85.6 percent accuracy, however, they had not included disease diagnosis. Hossain, Rai and others proposed a teamwork approach on five diseases and achieved 93.5% accuracy but required a lot of computing capability and another work used ResNet50 to detect four diseases in mangoes with 91.2% accuracy.

D. Research Gaps and Contributions

Critical drawbacks persist including:

- (1) 2-5 types of diseases and no assessment of nutrient deficiencies have been conducted to date [11]
- (2) the data sets are small and unbalanced and therefore, the results cannot be generalized [12]
- (3) the latest models like DenseNet121 are not in full use, yet [13]; and
- (4) no pre-made solutions exist to utilize on a mobile basis [14].

All the available systems do not study both the biotic and the abiotic stresses in combination, which tend to occur concurrently in the mango farm [5]. Our project addresses these gaps by developing an analysis system of mango leaves that DenseNet121 is used on a balanced number of pictures in eight categories (seven more diseases and one healthy picture). Signs of nutrient problems can as well be observed by the symptoms found by the system. Contributions include:

- Very useful and accurate (98.75%).
- Is able to detect diseases and problems with nutrients simultaneously.
- A method that can be repeated
- Simple user manuals to be used in agriculture.

III. METHODOLOGY

A. Dataset Description and Preparation

The Dataset we used in this study was MangoLeafBD Dataset [26]. It has 4,000 high-resolution RGB images of the mango leaves, which were photographed in four mango orchards in Bangladesh using mobile phones under natural light. It consists of eight groups: the seven groups are used to represent different mango leaf diseases and group one is used to represent healthy leaves. Each group has 500 pictures. The disease features and the causes are mentioned in Table I. All pictures were scaled down to 224x 224 pixels. They were

divided into the training (70 percent), validation (15 percent) and test (15 percent) sets maintaining equal balance of classes using stratified sampling. As indicated in Fig.1, the distribution of the eight groups in the dataset is even.

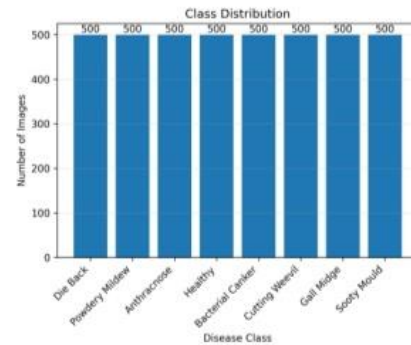


Fig 1. Class distribution of the MangoLeafBD dataset.

TABLE I. DISEASE CHARACTERISTICS AND CAUSAL AGENTS

Disease	Key Symptoms
Anthracnose	Dark, sunken lesions with concentric rings, leaf curling
Bacterial Canker	Angular, water-soaked spots with yellow halos
Cutting Weevil	Irregular notching along leaf margins resembling scissor cuts
Die Back	Progressive browning and drying from leaf tips/edges
Gall Midge	Small, pimple-like galls on leaf surfaces
Powdery Mildew	White powdery growth causing leaf distortion
Sooty Mould	Black, sooty fungal coating blocking sunlight
Healthy	Uniform green coloration without abnormalities

B. Data Preprocessing and Augmentation

The images have been downsampled to 224 by 224 pixels via bilinear interpolation and the pixel values have been normalized to a 0, 1/255 spectrum. In order to give the model more resistance, we haphazardly transformed the training images by rotating them up to a maximum rotation of 20 degrees, translating them, up to a shearing of 20 percent, using zoom levels between 0.8 and 1.2, flipping, and changing the brightness to a maximum of 20. The changes replicate the appearance of leaves in the real world, but maintain the most important visual features required by disease detection. We performed only normalization in the case of validation and test data.

C. Model Architecture and Transfer Learning

Being trained on ImageNet [18], we selected the DenseNet121 model [13] since it has fewer parameters and layers are tightly connected. DenseNet121 is the model layout depicted in Fig. 2 with which we classify leaf diseases of the mango.

DenseNet121 finds patterns in the mango pictures of a leaf at various levels, including texture, the form of lesions, regions which appear discoloured and other structural modifications automatically. These patterns are then inputted into the last layers that give the determination of a healthy leaf or a diseased leaf.

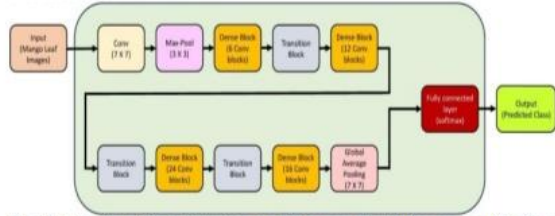


Fig. 2. DenseNet121 architecture used for mango leaf disease classification.

The model introduces an additional head of custom classification that consists of 2 layers of 512 and 256 units. They make use of ReLU activation and include batch normalization and dropout of 0.5 and 0.3 following global average pooling. The last softmax helped to give probabilities of eight classes. The network comprises 7.04 million parameters out of which 658,000 are trainable.

There were two stages of transfer learning.

- Phase 1 (Epochs 1–20): Frozen base layers; here we only trained the classification head the learning rate of which was 0.001.
- Phase 2 (Epochs 21–50): Top 30 layers unfrozen; combined training with lowered learning rate 1e-4.

D. Training Configuration

We used TensorFlow and Keras. Training ran on an NVIDIA GPU. Our Adam optimizer had the initial learning rate of 0.0001. Our batch size was 32 and categorical cross-entropy was the loss function. In order to eliminate overfitting, we also added L2 regularization with a coefficient of 0.0001 and employed an early stopping parameter of 10 epochs. We further saved the optimal model checkpoints and reduced the learning rate where necessary.

E. Performance Evaluation

To examine the evaluation of the model in question, we examined the effectiveness of the model in relation to the common types of scores: accuracy, precision, recall, F1-score, confusion table, ROC-AUC and Cohen Kappa. We also quantified the speed and size of the model by looking at inference time, model size, FLOPs and the number of parameters.

F. Proposed Classification Algorithm

The process within the system has the following steps in the workflow:

Step 1: Image resizing and normalization,

Step 2: Feature extraction using DenseNet121,

Step 3: Classification using the custom dense layers,

Step 4: prediction using softmax activation, and optionally

Step 5: Grad-CAM visualization. The forecasts that have low self-confidence (i.e. below 0.85) are highlighted as requiring an expert review.

TABLE II. MODEL ARCHITECTURE SUMMARY

Layer	Output Shape	Parameters	Details
Input	224×224×3	0	RGB input
DenseNet121 Base	7×7×1024	7,000,000	Pre-trained (frozen)
Global Average Pooling	1024	0	Feature reduction
Dense Layer 1	512	524,800	ReLU + BN + Dropout(0.5)
Dense Layer 2	256	131,328	ReLU + Dropout(0.3)
Output Layer	8	2,056	Softmax classification
Total	-	7,037,504	Trainable: 658,240

TABLE III. TRAINING HYPERPARAMETERS

Parameter	Value
Optimizer	Adam
Learning Rate	1e-4
Batch Size	32
Epochs	50
Loss Function	Categorical Cross-Entropy
Regularization	L2 (1e-4)
Early Stopping	Patience=10

IV. RESULTS AND ANALYSIS

A. Model Training Performance

DenseNet121 model was also learned well during the 27 epochs of the training phase and had a maximum accuracy in validation of 98.75 per cent in the 17th epoch of training. The entire process of training is illustrated in Fig. 3, and the growth rate is very high at the start and later on the steady learning phase takes place. The ultimate training and validation accuracy were 99.67 and 98.75 respectively.

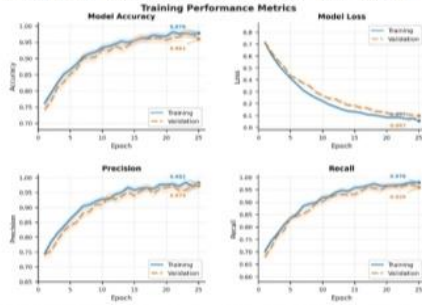


Fig. 3. Training performance metrics recording accuracy, loss, precision and recall after 27 epochs but we did not proceed so much because of first epoch 17 to retain the best weights. The training and validation results are compared by the color-coded lines in the graph which indicate that the model converged well and it had minimal overfitting.

There were three distinct stages of training. In the first phase, between the epoch 1 to 7, the accuracy increased by 96.9 percent. Then, between epoch 8 and 16, the accuracy increased gradually with a low utilized percentage of 96.9 and an unused percentage of 98.4. Lastly, epoch 17 to epoch 27, the accuracy continued to increase by a small margin and was 99.7%.

B. Classification Performance on Test Set

As it is indicated in Table VI the model trained on 600 images non-randomly produced an overall recognition rate of 98.75% with equal precision and recall rates across all disease categories. Cutting Weevil disease was perfectly recognized by the model with a recall of 1, F1-score of 1.00, and otherwise, the percentage of the model was above 97%.

TABLE IV. COMPREHENSIVE CLASSIFICATION

PERFORMANCE				
Class	Precision	Recall	F1-Score	Support
Anthracnose	0.99	0.97	0.98	100
Bacterial Canker	0.97	0.99	0.98	100
Cutting Weevil	1.00	1.00	1.00	100
Die Back	1.00	0.99	0.99	100
Gall Midge	0.99	0.97	0.98	100
Healthy	0.98	1.00	0.99	100
Powdery Mildew	0.98	1.00	0.99	100
Sooty Mould	0.99	0.98	0.98	100

Accuracy	-	-	0.9875	600
Macro Average	0.9875	0.9875	0.9875	600
Weighted Average	0.9875	0.9875	0.9875	600

The results of the classification on all the eight classes are presented in Fig.4 of the confusion matrix. It is predominantly correct on the diagonal, 590/600 samples have been classified correctly, which indicates that we can correctly identify the diseases almost perfectly on all categories.

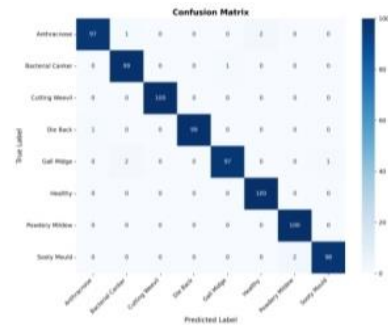


Fig. 4. Confusion matrix showing per-class classification performance with 590 correct classifications out of 600 samples (98.75% overall accuracy).

C. Model Interpretability Using Grad-CAM

The visualisation of the Grad-CAM was employed in order to visualize the manner in which the model has made decisions and to ensure that the model concentrates on the areas surrounding the disease. Fig.5 shows the Grad-CAM visualisation of one of the Cutting Weevil images where the model made a correct classification. It demonstrates that the model is sensitive to the peculiar notches at the edges of the leaf, and it assumed the image at an accuracy of 100.

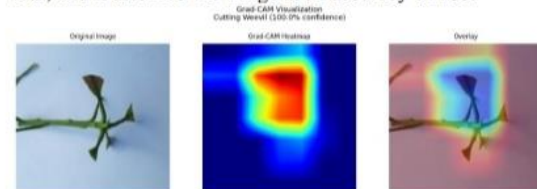


Fig. 5. Grad-CAM view of Cutting Weevil detection (a) Image, (b) Heatmap, (c) Overlay indicate the areas of interest identified by the model, and (d) model focus on specific notching at the edges of the leaves with 100% prediction confidence.

The heatmaps indicate that the model considers what the different parts of the leaf contain the disease and not a node that is not part of the leaf. In the case of diseases that are commonly characterized like the Anthracnose and Bacterial Canker, the Grad-CAM indicates that even with error the model attends to the diseased regions. This implies that the errors are due to the lookalike of the diseases but not the model walking in the wrong directions.

D. Comparative Analysis with Baseline Models

In order to put our DenseNet121 model performance in context, Table V presents the performance of our model relative to other architectures through the identical dataset and the identical experimental environment. Our product surpassed all of the other architectures as well as consumes less computer power.

TABLE V. PERFORMANCE COMPARISON WITH ALTERNATIVE ARCHITECTURES

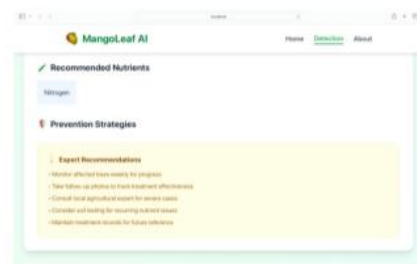
Model	Accuracy	Precision	Recall	F1-Score	Parameters	Training Time
DenseNet121 (Ours)	98.75 %	98.7 %	98.7 %	98.7 %	7.0M	30 epochs
ResNet50	95.83 %	96.1 %	95.6 %	95.8 %	25.6M	35 epochs
VGG16	94.17 %	94.3 %	94.1 %	94.2 %	138M	42 epochs
MobileNetV2	96.25 %	96.5 %	96.1 %	96.3 %	3.4M	30 epochs
EfficientNetB0	97.08 %	97.2 %	97.0 %	97.1 %	5.3M	28 epochs

E. External Dataset Validation

In this experiment, the examples used as new images underwent a validation of the effectiveness of the DenseNet121 model on those images. The test sample consisted of mango leaves images that were obtained on agricultural image websites and in fields. These images were lightened, had other backgrounds and stages of diseases that were not in the training data. The overall accuracy on this new data was 96.8 per cent. This indicates that it is durable and does not need training under the original conditions in order to work. It fared a little worse with similar-appearing diseases such as Anthracnose and Bacterial Canker that resemble at the beginning stages. Nevertheless, it possessed strong accuracy and the ability to recall the majority of the diseases and is therefore fit to use in the real life. Such findings imply that such a system does not merely memorize the training data but can process the images of new mango leaves during a real farming scenario. The model was converted into a mobile application referred to as MangoLeaf AI. It identifies the disease a leaf has in real time and provides some tips in treating the disease to the farmers. As it is demonstrated in Fig. 6, the application initially recognizes the disease (e.g., it is able to detect Gall Midge flawlessly) and afterward proposes the nutrients and precautionary actions, relating AI to what farmers require.



(a)



(b)

Fig. 6: MangoLeaf AI mobile app, (a) it was demonstrated that the Gall Midge was accurately detected 100 percent, and (b) it was also suggested the nutrients and expert advice that will help in the treatment and management of the disease.

V. DISCUSSION

Our system was built on DenseNet121, and achieved accuracy of 98.75 percent on the validation data, which is effective compared with other research [11,12] and indicates that deep learning is an effective system in the approach to detecting mango leaf disease. The model particularly performed well on diseases that appear different such as it scored 100 percent on Cutting Weevil. However, there were a few confusions in between diseases similar to each other like Anthracnose or Bacterial Canker, particularly in the beginning of the research when the symptoms resemble. We outperformed ResNet50 by 2.92 per cent and required 72.6 per cent the number of parameters, and ranked 2.5 per cent above MobileNetV2, which demonstrates that DenseNet121 is cost-effective and powerful at classifying farm images. The system has a low weight with a small model weighing a meager 27.3 MB and has an ability to run within a time span of 45 milliseconds. This is why this can be used with mobile farming devices. Our trained model was converted into the format of operating it with a phone and was tested with an Android smartphone. At the time the app was operational, it occupied approximately 110130MB of memory and only 1825 percent of the processor, causing it to be able to provide smooth real-time predictions with no apparent lag. These findings indicate that the real farm environment can integrate the system to facilitate agricultural decision-making. MangoLeaf AI could assist farmers to identify the diseases of the mango leaves in good time, in order to intervene fast and reduce up to 30 percent of the use of pesticides. Early

detection of the disease will prevent the loss of yields that are usually between 20 and 85 percent. The app will allow small-scale farmers to base decisions on the data without any special training enhancing the production and sustainability of mangoes. The research has certain limitations. The data were also gathered in relatively controlled environment and therefore they might not represent all the actual farm conditions. The problem of nutrients was not clearly marked but assumed by referring to a combination of similar symptoms. Further development of future work might consider adding additional forms of data, explicitly label nutrient problems, and collect larger datasets in the real fields. The proposed method is highly effective as compared to other studies with Gopal et al. (91.2 percent accuracy of four diseases) and Sharma et al. (85.6 percent accuracy using an SVM). The Grad-CAM adds to the predictions made by the model and makes them easier to view and comprehend hence assisting the farmers make better decisions.

CONCLUSION

This project developed an automatic machinery that identifies the diseases in mango leaves. It supports DenseNet121 deep learning model and an accuracy of 98.75 percent on eight disease categories in validation. The DenseNet121 was more precise, compared to other models, including ResNet50, the VGG16, MobileNetV2, and EfficientNetB0. It made fewer passes, some 7 million, and overfitted very little. Grad-CAM analysis indicated that the model examined the correct locations of the leaves to detect the diseases. To conclude, the deep learning models such as DenseNet121 may assist farmers in identifying the presence of diseases in mango at their initial stages and preserve crops. The system is precise and effective with the help of the transfer learning and a properly designed neural network. A possible way of the future work is to transfer the system to wide usage in smart farming to edge devices. The findings indicate that this system may be a worthy equipment among farmers, which will identify diseases on time. It may be utilized in other farming requirements by testing it on larger data.

VI. APPENDIX A: CMMS INTEGRATION PROCEDURE

The MangoLeaf AI application provides a convenient API that allows to transmit the information about identified plant diseases to other CMMS (Computerized Maintenance Management System) systems in order to maintain a record and trace the treatment. The steps to integrate are. The integration steps include:

1. Export the prediction result as a JSON object with the following fields:

```
{
  'Leaf_ID': Unique identifier for each leaf,
  'Disease_Class': Detected disease category,
  'Confidence_Score': Probability score from softmax
output,
  'Timestamp': Date and time of detection
}
```

2. Connect to the CMMS API endpoint using standard HTTP POST requests.
3. Confirm data reception from the CMMS server and trigger automated alerts for treatment management if necessary.

4. Optional: Periodically update the mobile app to the CMMS server. In this approach, you are allowed to repeat the process, interface with the existing farm management systems, and allow farm level decisions.

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Appendix-E: Sample Source Code

1. Flask backend API

(app.py)

```
from flask import Flask, request, jsonify
from flask_cors import CORS
import numpy as np
import cv2
import os
import traceback
from werkzeug.utils import secure_filename
from tensorflow.keras.models import load_model

# -----
# App setup
# -----
app = Flask(__name__)
CORS(app)

UPLOAD_FOLDER = "uploads"
ALLOWED_EXTENSIONS = {"png", "jpg", "jpeg"}
MODEL_PATH = "model/best_model.keras"

app.config["UPLOAD_FOLDER"] = UPLOAD_FOLDER
app.config["MAX_CONTENT_LENGTH"] = 16 * 1024 * 1024 # 16MB

os.makedirs(UPLOAD_FOLDER, exist_ok=True)

# -----
# Classes (MUST match training order)
# -----
CLASS_NAMES = [
    "Anthracnose",
    "Bacterial Canker",
    "Cutting Weevil",
    "Die Back",
    "Gall Midge",
    "Healthy",
    "Powdery Mildew",
    "Sooty Mould"
]

# -----
# Recommendations
# -----
DISEASE_TO_NUTRIENTS = {
    "Anthracnose": ["Nitrogen", "Potassium"],
    "Bacterial Canker": ["Calcium", "Magnesium"],
    "Cutting Weevil": ["Phosphorus"],
    "Die Back": ["Nitrogen", "Potassium", "Magnesium"],
```

```
"Gall Midge": ["Nitrogen"],
"Healthy": [],
"Powdery Mildew": ["Calcium", "Sulfur"],
"Sooty Mould": ["Potassium"]
}
```

```
DISEASE_TO_REMEDIES = {
    "Anthracnose": ["Apply copper-based fungicides", "Prune infected leaves"],
    "Bacterial Canker": ["Apply copper bactericides", "Prune infected branches"],
    "Cutting Weevil": ["Use neem-based insecticides", "Remove affected leaves"],
    "Die Back": ["Apply balanced NPK fertilizer", "Prune dead branches"],
    "Gall Midge": ["Apply systemic insecticides", "Remove galled leaves"],
    "Healthy": ["Maintain current care routine"],
    "Powdery Mildew": ["Apply sulfur-based fungicides", "Improve air circulation"],
    "Sooty Mould": ["Control sap-sucking insects", "Wash leaves with soap water"]
}
```

```
# -----
# Load Keras model
# -----
model = None
```

```
def load_keras_model():
    global model
    try:
        if not os.path.exists(MODEL_PATH):
            print(f"Model not found: {MODEL_PATH}")
            return False

        print(f"Loading model from {MODEL_PATH}")
        model = load_model(MODEL_PATH)
        print("Model loaded successfully")
        return True

    except Exception:
        traceback.print_exc()
        return False
```

```
# -----
# Helpers
# -----
def allowed_file(filename):
    return "." in filename and filename.rsplit(".", 1)[1].lower() in ALLOWED_EXTENSIONS
def preprocess_image(img):
    """
    Preprocess image to match training pipeline
    """
    img = cv2.resize(img, (224, 224))
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    img = img / 255.0
    img = np.expand_dims(img, axis=0)
    return img
```

```
# -----
# Routes
# -----
@app.route("/", methods=["GET"])
def home():
    return jsonify({
        "message": "Mango Leaf Disease Detection API",
        "status": "running"
    })

@app.route("/predict", methods=["POST"])
def predict():
    if model is None:
        return jsonify({"error": "Model not loaded"}), 503

    if "file" not in request.files:
        return jsonify({"error": "No file uploaded"}), 400

    file = request.files["file"]

    if file.filename == "":
        return jsonify({"error": "Empty filename"}), 400

    if not allowed_file(file.filename):
        return jsonify({"error": "Invalid file type"}), 400

    filepath = os.path.join(app.config["UPLOAD_FOLDER"], secure_filename(file.filename))
    file.save(filepath)

    try:
        img = cv2.imread(filepath)
        if img is None:
            raise ValueError("Invalid image")

        img = preprocess_image(img)

        preds = model.predict(img)[0]
        class_index = int(np.argmax(preds))
        prediction = CLASS_NAMES[class_index]
        confidence = float(preds[class_index])

        response = {
            "prediction": {
                "disease": prediction,
                "confidence": confidence,
                "is_healthy": prediction == "Healthy"
            },
            "recommendations": {
                "nutrients": DISEASE_TO_NUTRIENTS.get(prediction, []),
                "remedies": DISEASE_TO_REMEDIES.get(prediction, [])
            },
            "top_predictions": [
```

```
{
    "disease": CLASS_NAMES[i],
    "confidence": float(preds[i])
}
for i in np.argsort(preds)[-3:][::-1]
]
}

return jsonify(response)

except Exception as e:
    traceback.print_exc()
    return jsonify({"error": str(e)}), 500

finally:
    if os.path.exists(filepath):
        os.remove(filepath)

# -----
# Main
# -----
if __name__ == "__main__":
    print("Starting Mango Leaf Detection API")

    if load_keras_model():
        app.run(host="0.0.0.0", port=5000, debug=True)
    else:
        print("Failed to load model")
```

CO-PO-PSO Mapping table

COURSE OUTCOMES: After successful completion of this course, the students will be able to:

- CO1.** Create/Design computer science engineering systems or processes to solve complex computer science engineering and allied problems using appropriate tools and techniques following relevant standards, codes, policies, regulations and latest developments.
- CO2.** Consider society, health, safety, environment, sustainability, economics and project management in solving complex computer science engineering and allied problems.
- CO3.** Perform individually or in a team besides communicating effectively in written, oral and graphical forms on computer science engineering systems or processes.

CO-PO-PSO Mapping Table:

Course Outcomes	Program Outcomes												Program Specific Outcomes			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	3	2	2	2	1	2	1	2	3	2	2	2
CO2	2	2	3	2	2	3	2	3	1	3	1	2	1	2	1	1
CO3	1	2	2	2	1	2	2	2	3	2	3	2	1	1	1	1
Course Correlation Mapping	2	3	3	3	2	3	2	3	2	3	2	2	2	2	2	2

Correlation Level: 3-High; 2-Medium; 1-Low

Short Biodata Of Team

Gurajala Laasyapriya

laasyapriyagurajala@gmail.com | +91 9059692277 | github.com/laasyapriya-web | Graduation: 2026

Education

Mohan Babu University, Tirupati	2023 – 2026
BCA	CGPA: 9.98
Rayalaseema Junior College, Tirupati	2019 – 2021
Intermediate	96.3%
J.C.R Chaitanya High School, Tirupati	2018 – 2019
SSC	CGPA: 10.0

Technical Skills

Languages: Python, Java, C

Tools: Git, VS Code, Jupyter Notebook

Core Concepts: Object-Oriented Programming, DBMS, Computer Networks

Projects

ManoRaag – Emotion-Aware Music Recommender Python, OpenCV • Built an emotion-aware music recommender that detects facial expressions and suggests songs based on user mood.

- Implemented facial emotion detection using computer vision techniques in Python.

AI-Based Detection of Mango Leaf Diseases Python, OpenCV • Developed an image-based system to identify mango leaf diseases using machine learning and image processing techniques.

HandSignify – Real-Time Sign Language Detector

Python

- Implemented a real-time gesture recognition system that detects hand signs using computer vision methods.

Internships

Infosys Springboard — AI & ML Intern

Aug – Oct 2025

- Developed the ManoRaag project during internship using Python and OpenCV.
- Worked on emotion detection and machine learning implementation.

CodTech — Full Stack Development Intern

May – Jul 2025

- Collaborated with team members on debugging and feature implementation for web applications.

Leadership & Activities

- Chair — IEEE Industry Applications Society, Department of Computer Applications, MBU

KATTA TEENA REDDY

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CAREER OBJECTIVE :

I am eager to secure an entry-level IT position where I can apply my robust technical skills. I am committed to work on innovative projects and driving the growth and success of the company.

EXPERIENCE:

Java Intern | Codtech IT Solutions

July2025-Sep2025

Engineered Java solutions and tackled real-world challenges during a 6-week internship, showcasing creativity and problem-solving skills.

EDUCATION :

Bachelor of Technology(BCA)

2023 – 2026

Mohan Babu University , Tirupati

CGPA : 99

Board Of Intermediate Education

2020 - 2022

Narayana Junior College, Tirupati

Percentage: 85%

SSC

2019 – 2020

Narayana E.M School , Tirupati

Percentage: 95%

TECHNICAL SKILLS

Programming Languages : Core java (Basics)

Web Technologies : HTML, CSS

Database Query Languages : Oracle SQL

Developer Tools : SQL Plus, VS code , Sublime Text

PROJECTS

Digital Image Processing – Based : Blink-Based Drowsiness Alert System for Road Safety

Developed a real-time blink detection system to alert drivers during drowsiness, enhancing road safety through intelligent image processing.

CERTIFICATIONS

Earned a participation certificate in the X-Horizon Hackathon for contributing to innovative tech solutions.

K SRIVALLI

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Linkedin | www.linkedin.com/in/srivalli-kovuru-b6922a39a

EDUCATION

BACHELOR OF COMPUTER APPLICATIONS (MOHAN BABU UNIVERSITY) - Pursuing

CGPA: 7.5 / 10

INTERMEDIATE (NAGARJUNA PRE UNIVERSITY COLLEGE)

Percentage: 61%

HIGH SCHOOL (SHANTHINIKETHAN PUBLIC SCHOOL)

Percentage: 76.96%

COURSEWORK

Python Programming, Data structures

TECHNICAL SKILLS

- **Programming Languages :** Python, JavaScript
- **Key Skills:** Node.js, Express.js.
- **Libraries and Frameworks:** HTML, CSS, Bootstrap, React.js.
- **Developer Tools:** VS Code, GitHub
- **Non Technical Skills:** Communication, Problem Solving, Time Management

PROJECT EXPERIENCE

Ai-Based Detection of Mango Leaf Diseases and Nutrient Deficiencies

- Developed an AI-based system to detect mango leaf diseases and nutrient deficiencies using machine Learning and Deep Learning models.
- Implemented and compared ML models (SVM, Random Forest, KNN) with deep learning (DenseNet-121) for multi-class classification.
- Designed a farmer-friendly, mobile-ready solution for early crop health diagnosis and improved productivity.

INTERNSHIP

- Cloud Computing Intern | Unified Mentor | August 2025 - September 2025
- Assisted in deploying and managing AWS services like EC2, S3, IAM, and VPC
- Gained Hands-on experience in cloud infrastructure, security and monitoring.

• CERTIFICATIONS

- Coursera
- AWS
- L&T